5 Culvert

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5.1 Introduction

This chapter provides design procedures for the hydraulic design of highway culverts which are based on the Federal Highway Administration (FHWA) publication, *Hydraulic Design of Highway Culverts* (FHWA 2012). This chapter focuses on the hydraulic design of culverts that convey water. Culverts designed for purposes other than conveyance of water, like grade separated crossing for pedestrian and bicycle traffic are not covered in this chapter.

5.1.1 Culvert Definition

A culvert is defined as a conduit sized hydraulically to convey surface water runoff or stream flow under a highway, roadway, railroad, or other embankment. Culverts have the following characteristics:

- Buried structures covered with an embankment and generally composed of a structural material around the entire perimeter with some exceptions (e.g., bottomless culvert).
- Classified as a bridge when horizontal opening width is 10 feet or greater measured parallel to the roadway centerline. This includes multiple pipes where the clear distance between openings is less than half of the span of the smaller contiguous opening. See Section 5.2.2 for more information.
- Analyzed using the hydraulic design procedures defined in this chapter regardless of their classification.
- May include a structure; for example, median drains located in a divided highway are considered culverts whether they have a drop inlet or an apron on the upstream end, as long as they have an open outlet.

5.1.2 Concept Definitions

The following are descriptions of terms that are important in culvert design.

Allowable Headwater

The depth of water that can be ponded at the upstream end of the culvert during the design discharge.

Aquatic Organism Passage (AOP)

As used in this manual, the degree to which organisms living in the water can traverse a stream crossing such as a culvert, as compared to the adjacent stream channel. A principal, but not exclusive, component of AOP is fish passage.

Barrel Area

Cross-sectional area of a culvert measured perpendicular to the flow that is a function of the culvert dimensions. A larger area will convey more flow.

Barrel Length

The total culvert length used in hydraulic design is measured from the entrance to the exit of the culvert. This is typically based on where the crown intersects the fill slope. Because the height of the barrel, barrel slope, and barrel skew influence the actual length, an approximation of barrel length is usually necessary to begin the design process. This design length may differ from measured lengths used in construction plans and specifications.

Barrel Roughness

Barrel Roughness is a function of the material used to fabricate the barrel. Typical materials include concrete, corrugated metal, and plastic. The roughness is represented by a hydraulic resistance coefficient such as the Manning's n value.

Barrel Slope

Barrel slope is the actual slope of the culvert barrel and is often the same as the natural stream slope.

Bottomless Culvert

Bottomless culverts are three-sided structures that have sides on footings, with a top, and use the natural channel for the bottom.

Bridge Culvert

Buried structures that convey water through an embankment with horizontal dimensions greater than or equal to 10 feet measured parallel to the roadway centerline. This includes multiple pipes where the clear distance between openings is less than one-half of the smaller contiguous opening.

Centerline Culvert

A culvert that crosses under a roadway centerline conveying water from one side of the roadway embankment to the other.

Check Storm

An event larger than the design storm used to evaluate the design by analyzing outcomes of events that exceed the design storm. The check storm varies depending on type structure, level of acceptable risk, historical events, and estimates of future events. When performing a hydraulic analysis with a check storm, the design is not expected to meet all design criteria, rather the designer estimates the flood impacts during extreme storms and considers resiliency improvements to reduce flood risk potential.

Critical Depth

The depth at which the specific energy of a given flow rate is at a minimum. For a given discharge and cross-section geometry there is only one critical depth.

Crown

The inside top of the culvert or pipe. It is also referred to as the soffit.

Culvert

A conduit pipe with open ends which conveys surface water under a road or embankment. A culvert may include a structure; for example, median drains located in a divided highway are considered culverts whether they have a drop inlet or an apron on the upstream end, as long as they have an open outlet.

Design Discharge

The rate of flow that a drainage facility is expected to accommodate without exceeding the design criteria.

Design Flood

A flood event having the design discharge and design frequency. This is also referred to as the design event.

Design Frequency

The recurrence interval of the design discharge. This is also referred to as the flood frequency.

Entrance Culvert

A culvert located on MnDOT right of way conveys water through an entrance embankment that allows access to public, commercial, or private property. The most common types of private entrances are agricultural (field or farm) or residential driveways which typically have less traffic.

Equivalent Diameter

Diameter of circular pipe with approximately the same cross-section area as the arch or elliptical shaped pipe of similar material.

Flowline, Culvert

The bottom invert of a conduit. When the invert is buried with riprap or other fill material placed in the culvert, the flowline is then the top of the fill material.

Flume

An inclined open or closed pipe or channel used to convey water down a slope. A flume holds water as a complete structure and can be natural or manmade.

Free Outlet

An outlet having a tailwater equal to or lower than critical depth. For culverts having free outlets, lowering the tailwater has no effect on the backwater profile upstream of the tailwater.

Frequency

The number of times a flow, precipitation event, or flood of a given magnitude or greater can be expected to occur on average over a long period of time. Frequency analysis is the estimation of peak discharges for various recurrence intervals. Another way to express frequency is with probability. Probability analysis seeks to define the flood flow with a probability of being equaled or exceeded in any year at a specific location.

Headwater (HW)

The water surface elevation on the upstream side of a culvert or bridge providing the energy to force water through the culvert or bridge opening. For bridges the headwater is typically measured at the approach section.

Improved Inlet

Flared, depressed, or tapered culvert inlet that decrease the amount of energy needed to pass the flow through the inlet and thus increase the capacity of culverts with inlet control or supercritical flow.

Inlet Control

One of two basic types of flow control in culvert hydraulics where the culvert barrel is capable of conveying more flow than the inlet will accept.

Invert

The inside bottom of the pipe or culvert.

Major Culvert

A round culvert diameter, arch or elliptical culvert equivalent diameter, or box culvert span greater than 48 inches. A multiple barrel culvert with a combined area exceeding the area of a 48 inch round pipe is also considered a major culvert.

Median Drain

A culvert conveying water from a median inlet, either an apron or drop inlet, to an outlet on the other side of a road embankment.

Minor Culvert

A round culvert diameter, arch or elliptical culvert equivalent diameter, or a box culvert span of 48 inches or less. A multiple barrel culvert with a combined area equal to or less than an the area of a 48 inch round pipe is also considered a minor culvert.

Non-roadway Culvert

Culvert that does not go under a roadway embankment. These culverts may go through ditch blocks or check dams; or under bike, pedestrian, or snowmobile trails.

Normal Flow

A flow condition where the water surface slope and energy grade line are parallel to the bed slope. This occurs in a channel reach when the water velocity, depth, and slope remain constant along the reach. This type of flow will exist in a culvert operating on a uniform slope when the culvert is sufficiently long.

Outlet Control

One of two types of flow control in culvert hydraulics where the barrel is not capable of conveying as much flow as the inlet opening will accept.

Overtopping

Flow of water over the top of a road, embankment, or dam. The most likely location of roadway overtopping is at the sag (low point) in the road profile, which may be offset from the culvert or bridge location.

Overtopping Frequency

The probability that an overtopping flood event will be equaled or exceeded in any year at a specific location. This is also referred to as the overtopping flood.

Point of Intersection (P.I.)

Location where two lines meet. The shoulder P.I. is the location where the shoulder meets the embankment side slope (inslope).

Side Culvert

A culvert on MnDOT right of way that conveys water under a local road or an entrance that intersects a trunk highway or other embankment. Side culverts are often parallel to the trunk highway.

Stage Increase

The difference between headwater and the unconstricted water surface just upstream of the culvert or bridge. This is also referred to as the backwater or swellhead.

Submergence

Occurs when the culvert inlet and/or the outlet goes under water. A submerged outlet occurs where the tailwater elevation is higher than the crown of the culvert. A submerged inlet occurs where the headwater is greater than 1.2 times the culvert diameter.

Tailwater (TW)

The depth of water downstream of a hydraulic structure. This measurement is taken at a location where the structure does not impact the water surface elevation.

5.2 Design Criteria

Culvert design criteria are the guidelines and standards that form the basis for the selection of the final design configuration. There are many different culvert sizes, shapes, and materials from which a designer can choose. The design selected should be the one that best integrates:

- hydraulic efficiency,
- serviceability,
- structural stability,
- economics,
- environmental considerations,
- resilience,
- traffic safety,
- constructability, and
- land use requirements.

Culverts are used in the following conditions:

- When more economical than a bridge
- Where bridges are not hydraulically required
- Where higher in-stream velocities can be tolerated
- Where greater stage (headwater or tailwater) increases can be tolerated
- Where debris and ice are tolerable (Section 5.2.5.3)

5.2.1 Design Frequency

Culverts are referred to as either a minor or major culvert. Minor culverts have a pipe diameter, arch or elliptical pipe equivalent diameter, or box culvert span of 48" or less, while major culverts have a pipe diameter, arch or elliptical pipe equivalent diameter, or box culvert span greater than 48". All steel structural plate culverts are major culverts. Equivalent diameters for concrete and metal pipes are provided in Chapter 2.

When a culvert consists of two or more pipes use the combined area of the pipes to determine the equivalent round pipe size. Use the equivalent round pipe diameter or sum of the box culvert span to determine if a multi-barrel culvert is a major or minor culvert.

The design frequency for centerline culverts, including interstates and highways, are provided in Table 5.1. Ramps, loops, rest areas, and some side culverts will also use the centerline design frequency criteria. In some situations, such as long highway routes without practical detours, critical routes, deep fills, or the replacement of culverts damaged by flooding, use of a larger design frequency may be justified. Local roads with lower Annual Average Daily Traffic (AADT) may consider use of a smaller, more frequent, design event.

Table 5.1 Design Frequency Recommendation by Road and Culvert Classification

Road and culvert classification	Exceedance probability (percent)	Design frequency
All centerline > 48 inches (major culvert)	Varies	Complete a Risk Assessment
All centerline ≤ 48 inches (minor culvert)	2%	50-year
Median drain	2%	50-year
Entrance	10%	10-year

Median drains, entrance culverts and non-roadway culverts typically will not require a larger design event or a Risk Assessment (See Appendix A). However, the designer should evaluate the site to determine if a larger design frequency or a check storm analysis is needed.

Entrance culverts are sometimes designed with an overtopping area to handle flood events greater than the design event. Standard Plate 9000 provides recommended standards for approaches and entrances when site conditions allow overtopping.

Non-roadway culverts include those used for pedestrian, bike, and snowmobile trails. Design frequency selection for non-roadway culverts is site-specific but may range between a 2-year event through a 50-year event. Select the design frequency based on type and amount of use, location, site conditions, cost and acceptable flood risk associated with the culvert.

Determination of the culvert size is an iterative process. Data collection is an important first step in assessing site conditions, historical flooding, drainage issues, and past performance of existing culverts. In most cases the existing culvert is used as the initial size unless there is evidence that the culvert is undersized or oversized. Design procedures for selecting culvert sizes are provided in Section 5.4.

5.2.1.1 Minor Culvert Design Frequency

Use the 50-year design frequency for most minor culverts. The overtopping frequency does not need to be computed; however, the roadway should not overtop during the design frequency.

A minimum design frequency of 10-years may be used for lower risk entrance culverts. Check that the adjacent roadway does not flood during the roadway minimum overtopping frequency. The entrance may need to be designed with an overtopping area to accommodate larger flood events.

A lower design frequency such as a 25-year event can be considered for local road or side culverts where the projected AADT is less than 1500 and there is a low risk of flood damage or impacts to traveling public. Considerations include availability of a detour route, traveler expectations based on road classification, flood damage potential, history of road or culvert flood damage, and design criteria.

A higher design frequency, such as a 100-year design event, may be required if there is significant flood damage potential upstream, history of overtopping, special traffic considerations (such as emergency routes, access to critical locations), or to accommodate FEMA mapped floodplain regulations.

5.2.1.2 Major Culvert Design Frequency

Major culverts are evaluated for a range of design discharges from the 2-year to the 500-year design frequency, or the overtopping flood if it is less than 500-year event. A Risk Assessment Form has been developed by MnDOT to document the considerations given to flood risk including backwater damage potential, traffic related losses, and roadway and/or structure repair costs for bridges and culverts (see Appendix A). This form should be applied to all waterway bridges (see Chapter 9) and major culverts to evaluate whether the proposed design results in an acceptable flood risk.

Minimum overtopping flood frequency used in the risk assessment is based on projected AADT. Since overtopping should not occur during the design frequency, the roadway should not be overtopped during the design discharge. In some situations, designing for an overtopping event that is greater than the minimum listed in Table 5.2 is justified. Overtopping during the design discharge is limited by the allowable headwater design criteria (See Section 5.2.3.1).

Table 5.2 Minimum Overtopping Frequency Based on Projected Annual Average Daily Traffic (AADT)

Projected AADT	**Minimum overtopping flood frequency**
0 - 10	2-year
10 - 49	5-year
50 - 399	10-year
400 - 1499	25-year
1500 and up	50-year

5.2.2 Bridge Culverts

Minnesota Statute 165 defines a bridge “as a structure, including supports erected over a depression or an obstruction, such as water, a highway, or a railway, having a track or passageway for carrying traffic or other moving loads, and having an opening measured horizontally along the center of the roadway of ten feet or more between undercopings of abutments, between the spring line of arches, or between the extreme ends of openings for multiple boxes. Bridge also includes multiple pipes where the clear distance between openings is less than one-half of the smaller contiguous opening.”

The *LRFD Bridge Design Manual* identifies culverts as buried structures. Buried structures with horizontal dimensions greater than or equal to 10’0” measured parallel to the roadway centerline are considered bridges. This includes multiple pipes where the clear distance between openings is less than one-half of the smaller contiguous opening. Figure 5.1a shows how clear distances are computed for multi-cell box culverts. Figure 5.1b shows how clear distances are computed for multiple culvert barrels and applies to any shape culvert including round, arch, and box culverts.

Bridge culverts are given a bridge number and are entered into MnDOT's bridge inventory system. All box culverts require a plan prepared by the Bridge Office or a qualified structural engineer. Round and arch pipe sizes listed on the standard plates are structurally designed and can be used within the minimum cover and maximum fill height criteria in Chapter 2. Sizes not included in the Standard Plates or not being used within the constraints of the Chapter 2 load tables will require a structural design; consult the Bridge Office. Additional information on the hydraulic design of bridges over water is provided in Chapter 9.

5.2.2.1 Bridge Culvert Opening

Dimensions used to determine if a culvert is classified as a bridge are shown in Figures 5.1a, 5.1b, and 5.2. The bridge opening is not measured perpendicular to the barrel wall (pipe diameter or span) but is measured parallel to the roadway centerline. The bridge opening includes both the length of the barrel openings and the clear distance between openings divided by the cosine of the angle of skew between barrel and roadway centerline. The bridge opening will not necessarily be the same as the dimension used for hydraulic design of culverts.

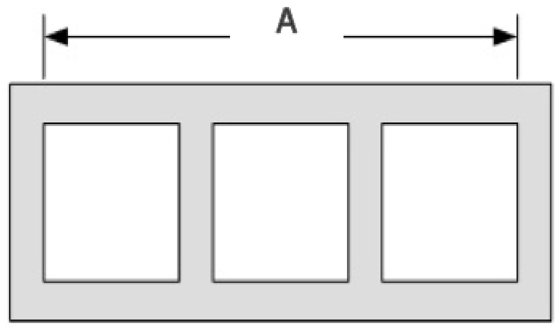


Figure 5.1a Profile View of a Four-Sided, Multi-Cell Culvert Under Fill
Source: *Specifications for the National Bridge Inventory* (FHWA 2022)

The profile view of a multi-barrel culvert Figure 5.1b provides an example with two barrels that have a clear distance less than half of the smaller contiguous opening (barrel d1) and are considered to both be part of the bridge culvert. The third barrel has a clear distance equal to or greater than half the smaller pipe opening (barrel d2) so it is not considered to be part of the bridge culvert. The guidance applies to any shape culvert including round, arch, and box culverts.

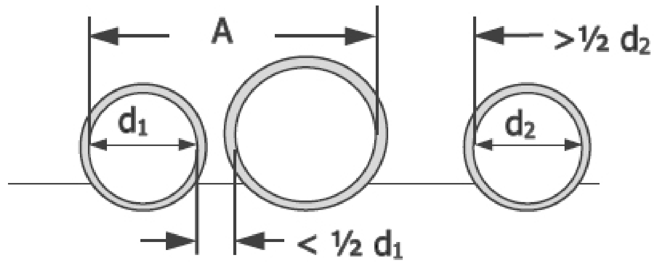


Figure 5.1b Profile View of a Multi-Pipe Culvert Under Fill
Source: *Specifications for the National Bridge Inventory* (FHWA 2022)

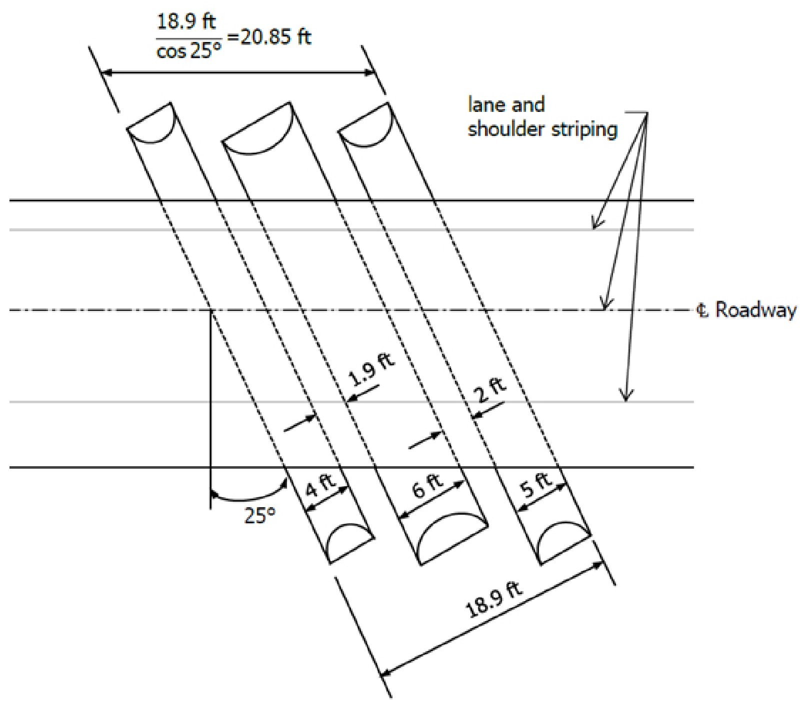


Figure 5.2 Example Plan View of a Skewed, Multi-Pipe Culvert Under Fill
Source: *Specifications for the National Bridge Inventory* (FHWA 2022)

The example plan view in Figure 5.2 has three skewed barrels that all have a clear distance less than half the opening of the smaller contiguous openings, so all three pipes are part of the bridge culvert. Equation 5.1 is for a three-barrel culvert where distance between all culvert barrels is less than half of the smaller contiguous opening. The bridge opening is also considered to be the “bridge length” when determining if the culverts qualify as a bridge structure or not.

$$\text{Bridge Opening} = \frac{A}{\cos\Delta} = \frac{d_1 + d_2 + d_3 + c_1 + c_2}{\cos\Delta}$$ **Equation 5.1**

Example in Figure 5.2:

$$\text{Bridge Opening} = \frac{18.9'}{\cos 25^\circ} = \frac{4' + 6' + 5' + 1.9' + 2'}{\cos 25^\circ} = 20.85'$$

Where:

Bridge Opening is measured along the center of the roadway.
A = sum of pipe diameter and clear distance measured perpendicular to pipe
d_1, d_2, d_3 = pipe diameter or span
c_1 and c_2 are clear distance between openings of contiguous barrels, where $c_1 < 1/2\ d_1$ and $c_2 < 1/2\ d_3$
Δ = angle of skew in degrees between pipe and roadway centerline. When pipe is not skewed the angle is zero.

5.2.2.2 Multi-Barrel Bridge Culverts

In some situations, it may be desirable to have a bridge culvert with multiple barrels. A flow chart, Figure 5.3, has been developed to provide guidance on considerations in determining if a bridge, single span culvert, or multi-barrel culvert should be used. Use this chart after completing an initial hydraulic analysis when you are considering using multiple box culverts versus bridge-type structures. The flow chart can also be used for multi-barrel culverts with arch or round barrel shapes. This procedure is recommended for Trunk Highway bridge structures, though local agencies may also use it with additional criteria as appropriate.

Guidelines for determining the appropriate number of barrels are provided.

1. Using the flowchart, determine if a multiple barrel culvert (anything with more than one barrel) is feasible.

2. May use up to a 48-foot total open span with a maximum of three culvert barrels. Site-specific design considerations:

   - Recess the center (or one) of the culvert barrel to create a natural bottom.
   - Place the outer barrels(s) with the inverts at the overbank floodplain elevation, or similar.
   - If possible, span the bankfull width of the channel with a single culvert barrel.
   - Determine if flared end sections will be required to better channelize flow.
   - Culvert barrels will typically be parallel.

3. Anything above a 48-foot total open span and/or three barrels should only be considered if the following is completed by the District (or Local agency):

   - A cost analysis of both the multiple barrel culvert and a more typical bridge scenario.

- A more robust hydraulic analysis such as physical modeling or computational fluid dynamics (CFD). Standard HY-8 and HEC-RAS runs are generally limited to three barrels based on the limit of flume studies performed with more than three barrels to verify model results.

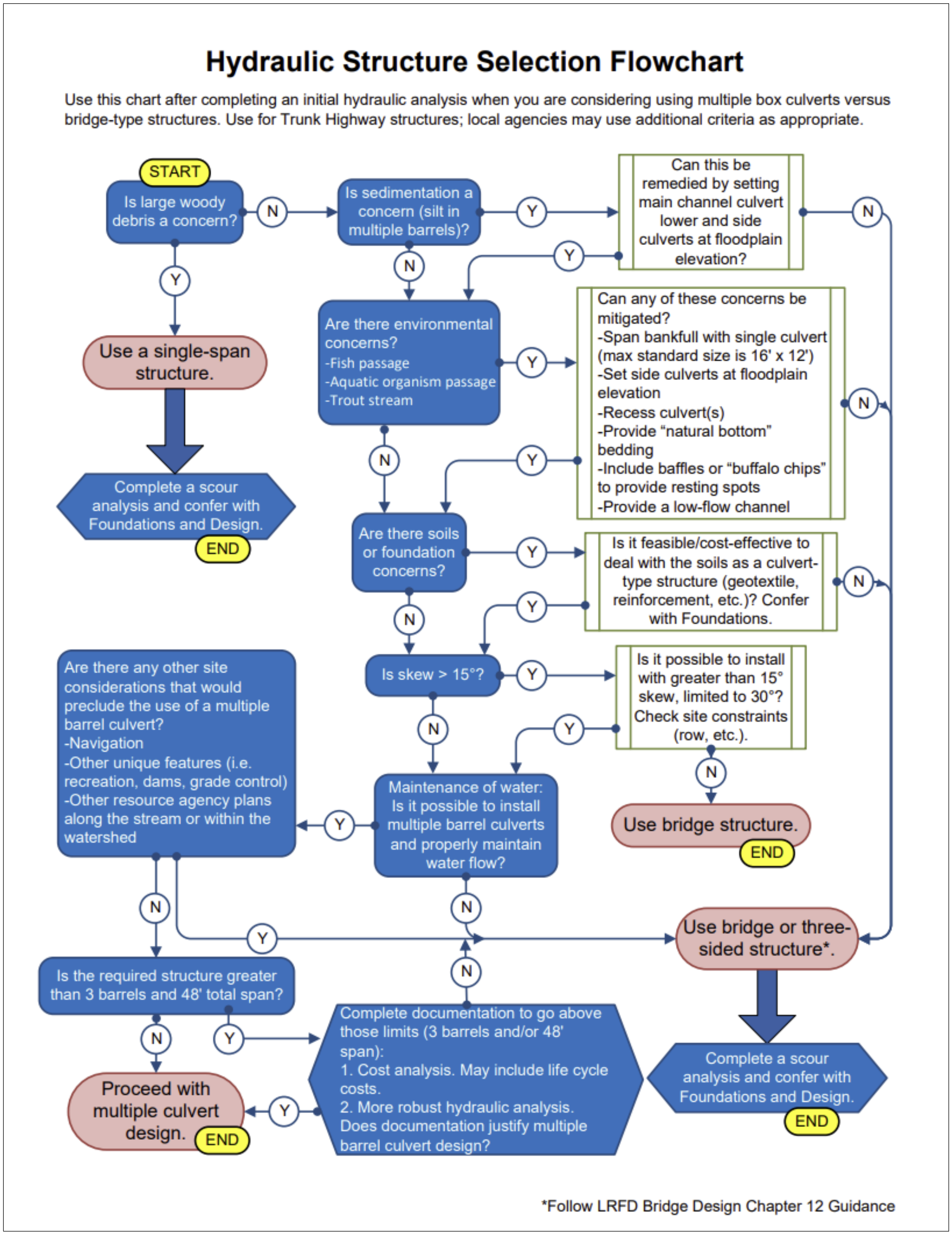


Figure 5.3 Hydraulic Structures Selection Flowchart

5.2.3 Design Criteria

There are several criteria that place limitations on the design of a culvert: allowable headwater, upstream storage, tailwater relationship of both the channel and any confluence, and outlet velocity minimums and maximums.

5.2.3.1 Allowable Headwater

Allowable headwater is the depth of water that can be ponded at the upstream end of the culvert during the design discharge.

Design centerline culverts with a minimum 1-foot freeboard between the headwater elevation and the roadway shoulder point of intersection (P.I.).

Design entrance culverts so the entrance does not overtop at the entrance culvert design frequency and does not cause flooding of the adjacent roadway during the roadway overtopping frequency. This may involve designing the entrance with an overtopping area to handle flood events greater than the entrance culvert design event.

Roadway overtopping should not occur during the design event with the allowable headwater criteria. Allowable headwater may also be limited if one or more of the following criteria is not met:

- It is non-damaging to upstream property;
- It is non-damaging to the roadway and road embankment;
- It meets criteria to limit stage increase set forth by regulatory agencies;
- It does not result in flooding of the driving lanes;
- It does not cause disruption to traffic flow;
- It does not damage the culvert.

Culverts in locations with high fill or steep fill slopes can experience piping, floatation, scour, or embankment erosion. It may be necessary to limit headwater by upsizing culverts with high or steep fill slopes or installing mitigation measures. When the headwater-to-diameter ratio (HW/D) is greater than 1.5, evaluate the culvert to determine if outlet velocities or increases in ponding durations need to be addressed.

Significantly extending the period of time water is stored against the embankment can result in damage to the embankment and roadway. Chapter 13 of the MnDOT *Facility Design Guide* covers how infiltration of ponded water can damage highway infrastructure and, in some situations, create a risk to the traveling public. An FHWA Memorandum provides rationale why roadways should not be used as levees or flood control structures (FHWA 2008).

Regulatory Agency Permit Requirements may also influence the allowable headwater. The designer must determine whether floodplain or other permit regulations apply to the project, and if so, how the regulations impact the design criteria. In some situations, temporary drainage must also be designed to meet permit requirements. Additional information about temporary drainage for bridge and culvert work in public waters is available in Chapter 13 of the MnDOT *Facility Design Guide* and the Minnesota Department of Natural Resources (DNR) publication, *Best Practices for Meeting DNR General Public Waters Work Permit*.

5.2.3.2 Upstream Storage

If a culvert is downsized to take advantage of the potential storage behind the roadway embankment, the designer is placing an obstruction in the drainage way. In this situation it may be necessary to analyze smaller, more frequent flood events to assess impacts to crop lands and adjacent areas.

If storage is being contemplated upstream of the culvert to reduce the peak outflow through the culvert, consider:

- If allowable headwater criteria can be met
- Potential damage to the roadway embankment, pavement, or culvert due to increased frequency and duration of flooding
- Any negative impacts of saturating the roadway embankment if the embankment acts as a levee
- Limiting ponding in urban and rural areas to non-sensitive locations, existing ponding, or storage areas
- Restricting ponding to non-crop producing locations in rural areas
- Minimizing increases in flood duration of existing cropland to reduce crop damage
- Limiting the total area of flooding
- Long term maintenance needs to preserve storage volume (e.g., removing sediment and debris)
- Purchasing right of way or easements, as necessary, to ensure that the storage area will remain available for the life of the culvert

5.2.3.3 Tailwater

Evaluate the hydraulic conditions of the downstream channel to determine a tailwater depth for a range of discharges, which include the design discharge and, in some situations, the check storm discharge. The following provides designers with general steps for determining hydraulic conditions; consult Chapter 4 for additional information:

- Tailwater depth is measured from the culvert outlet invert.
- A single section analysis is usually adequate for culvert design. Backwater curves can be calculated to transfer the tailwater elevation from the cross-section to the culvert site.
- Utilize a step backwater method such as provided in computer applications to determine tailwater elevations at sensitive locations.

- Use the critical depth and approximate hydraulic grade line if the culvert outlet is operating with a free outfall. In this situation a tailwater depth equal to $(d_c + D)/2$ where d_c is the critical depth of flow in feet and D is the culvert diameter in feet may apply.
- Use the headwater elevation of any nearby downstream culvert or other control structure if it is greater than the channel depth.

Designers should also consider the effects on tailwater elevations from nearby water bodies and confluences:

- Evaluate the high water elevation that has the same frequency as the design flood if events are known to occur concurrently and are statistically dependent.
- If statistically independent, evaluate the joint probability of flood magnitudes and use a likely combination resulting in the greater tailwater depth. Tailwater Joint probability analysis is covered in Chapter 8.

5.2.3.4 Outlet Velocity

The maximum velocity at the culvert outlet should be consistent with the velocity in the natural channel, stream, or river, or be mitigated with outlet protection. In some situations, culvert outlet velocities can be reduced to prevent scour or erosion by modifying the culvert size, slope, or roughness. Outlet protection alternatives are covered in Section 5.2.4.6. In general, outlet velocities less than 6 feet per second (fps) will not require energy dissipation or protection. Chapter 6 provides additional information on energy dissipation when higher velocities are present.

Minor culverts should be designed to maintain a minimum self-cleaning velocity unless the culvert was designed for Aquatic Organism Passage (AOP). Use a minimum velocity of 2.5 fps for the 2-year flood frequency when the streambed material size is not known.

5.2.4 Design Features

Basic design features and considerations include culvert size and shape, number of barrels, cover requirements, material selection, end section for both inlet and outlet, improved inlets, safety aprons and grates, and performance curves. Select the culvert size and shape based on engineering and economic criteria related to site conditions.

Performance curves for a range of discharges are developed for major culverts to evaluate the hydraulic capacity and outlet velocity of a culvert for various headwater depths. These curves display the consequence of high flow rates at the site and provide a basis for evaluating flood hazard.

5.2.4.1 Minimum Size

All culverts should be hydraulically designed to provide adequate hydraulic capacity. However, land use requirements, AOP adaptations, resilience considerations, and debris or ice potential may dictate a larger or different barrel geometry than required for hydraulic design alone. During design, consider the potential that debris or ice could plug a culvert. Use the following minimum sizes to avoid maintenance problems and clogging:

Table 5.3 Minimum Culvert Size Based on Roadway Type

Type of road	Minimum size
Interstate and trunk highway centerline	24 inches
CSAH centerline	18 inches
Local road centerline	18 inches
Ramps, loops, rest area	18 inches
Side culverts	15 inches
Median drains	15 inches
Entrances	15 inches
Non-roadway	15 inches

5.2.4.2 Pipe Shape and Configuration

Numerous cross-sectional shapes are available for both closed conduit and open-bottom culverts. The most common closed conduit shapes are circular, box (rectangular), and pipe-arch. Shape selection is based on the cost of construction, the limitation on upstream water surface elevation, roadway embankment height, and hydraulic performance. These shapes can be constructed with embedment which is a depression below the streambed of both the inlet and outlet inverts. Round and box culverts are the most common shapes. Arch pipes are more expensive than round pipe but are used when minimum fill cannot be met with a round pipe, to get over or under an obstruction, or to improve flow characteristics such as reducing the outlet velocity.

Multiple barrel culverts should fit within the natural dominant channel, or with minor widening of the channel unless barrel flowline elevations are adjusted to match the channel and floodplain geometry. Widening the channel at a culvert to allow multiple barrels at the same elevation typically leads to conveyance loss due to sediment deposition in some of the barrels. Multiple barrels should be avoided where there is a high velocity approach flow, particularly if supercritical flow is expected. These sites require either a single barrel or special inlet treatment to avoid adverse hydraulic jump effects.

Spacing between barrels should be designed to have sufficient space to get adequate compaction between the barrels while not having so much space that the barrels no longer act hydraulically as a single culvert. Interior barrel dimensions are used in the hydraulic design computations since only the open area conveys water. Where feasible, barrel spacing and orientation should fit within the stream geomorphology and flow patterns. Inlet embankment erosion and debris build up between barrels is more likely if the barrels have a significant separation. In some locations inlet protection will be needed.

Box culvert standard plan sheets include riprap at the inlet but erosion can still occur if the barrel placement does not consider stream flow patterns.

Drainage law and permit requirements may also impact culvert design. Minnesota State Statute 103E defines public drainage systems and identifies the authority for administrating them. The drainage authority may specify a culvert flowline and minimum capacity needed for the drainage system; existing county and judicial ditch flow lines may be incorrect due to scour or sedimentation. Due diligence is needed to assure that highway structures are designed for the correct flow lines. Review of drainage ditch records and consultation with the ditch authority may be necessary. The ditch authority may be the County Engineer, the County Ditch Engineer, or the Watershed District Engineer depending on the circumstances. See Chapter 1 for additional information on judicial ditches and other regulations.

DNR permits often include requirements, like aquatic organism passage (AOP), that may influence culvert design particularly for culverts on public waters or floodplains. AOP requirements often affect the pipe size, shape, configuration, and bottom material of a culvert design. Most AOP culverts are embedded or recessed pipes and designed with natural streambed material placed inside the pipe so that the flowline matches the streambed profile. Bottomless culverts are discouraged unless placed into bedrock or a non-erodible material due to the higher risk of failure from scour during extreme flood events. Bottomless culverts will require a risk assessment, structural design, and scour prevention for events larger than the design event; consult the Bridge Office if considering a bottomless culvert.

Where fish passage is required, special treatment may be necessary to ensure adequate low flow water depths. A pool riffle design can be used or, alternatively, if there are multiple barrels one barrel can be lowered to maintain water depth for the main channel. Additional information on AOP design is provided in Section 5.2.6.

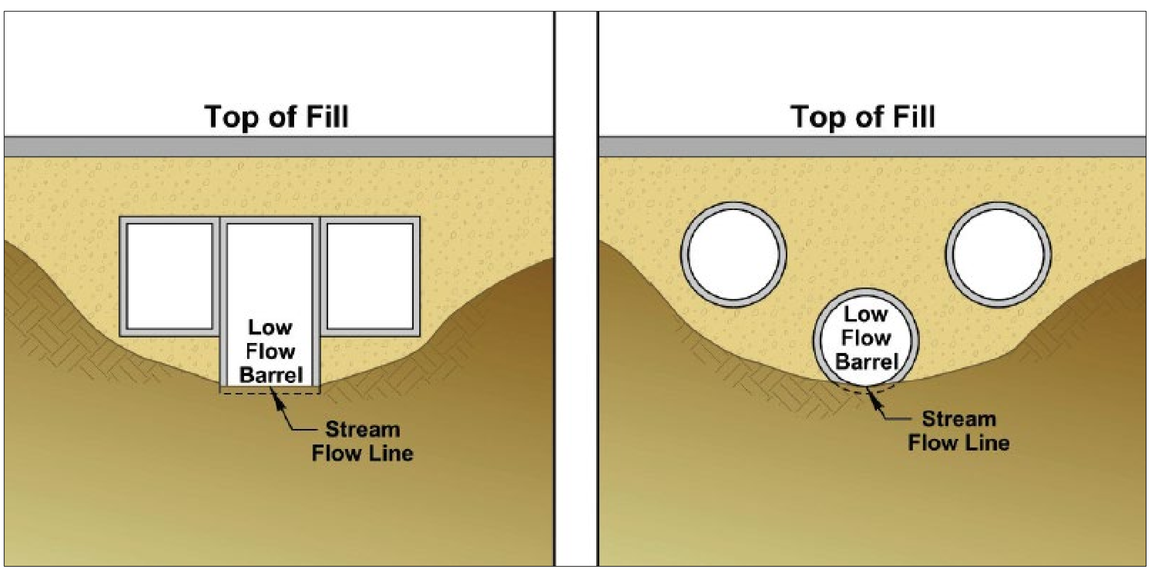


Figure 5.4 Multiple Barrel Culverts with One Low Flow Barrel
Source: *HDS-5, Figure 5.18* (FHWA 2012)

5.2.4.3 Pipe Material Selection

Barrel material selection is based on the material type that best fulfills all the engineering requirements for a specific installation. The following factors shall be considered: hydraulic performance, structural stability, serviceability, economics based on design life of structure, and replacement cost and difficulty of construction.

Abrasion and corrosion are also considered when determining serviceability requirements. The culvert design form (Figure 5.17) or other method of documentation should be provided for each centerline pipe installation indicating the engineering considerations that dictate the selection of the specific type of pipe. Guidance on pipe material service life is provided in Chapter 2.

5.2.4.4 Culvert End Sections

An apron end section is a concrete or metal structure attached to the end of a culvert for purposes of aesthetics, anchorage, and stabilization of the embankment near the waterway. Various end section designs are provided in the MnDOT Standard Plates, which are listed in Table 5.4 along with their associated entrance loss coefficients (k_e). The k_e values are used in entrance head loss calculations (see Section 5.3). Additional end section design options are listed in Table 5.6.

Table 5.4 Inlet and Outlet End Sections

Standard plate number	Standard plate description	Entrance Loss Coefficient, k_e
3022	Precast Concrete Safety Apron	0.7
3100	Concrete Apron for Reinforced Concrete Pipe	0.5
3110	Concrete Apron for Reinforced Concrete Pipe-Arch	0.5
3114	Sectional Concrete Apron for Reinforced Concrete Pipe-Arch	0.5
3122	Metal Apron for C.M. Pipe-Arch Culvert	0.5
3123	Metal Apron for C.S. Pipe	0.5
3125	Inlet Protection for Metal Culverts (90" dia. To 96" dia.)	0.2
3126	Inlet Protection for Structural Plate (60" thru 96" dia. or span)	0.7
3127	Inlet Protection for Structural Plate (102" thru 180" dia. or span)	0.7
3128	Metal Safety Apron & Grate	0.7
3129	Metal Apron for Corrugated Polyethylene Pipe (Use at Entrances and Driveways)	0.5
3148	Safety Slope Metal End Section for Circular & Arched Pipes	0.7

Standard plate number	Standard plate description	Entrance Loss Coefficient, k_e
Bridge Details Manual	Box Culverts	0.5

If culvert end sections are located within the clear zone there is a higher potential for vehicle impact. The designer must consider vehicle safety in end section design since some designs can be hazardous to errant vehicles. Designers should either place culvert ends outside the clear zone or use safety aprons.

- Culvert ends located outside the clear zone do not need safety aprons or grates.
- Culvert ends located within the clear zone should be treated in accordance with Chapter 13 of the MnDOT *Facility Design Guide*.
- When safety grates are required, assess the potential for flood damage if the grate plugs with debris or ice.

If a safety grate or trash rack is installed on the downstream end of a culvert, a safety grate or trash rack must also be installed at the upstream end of the culvert to prevent a person from becoming trapped due to intentional or accidental entry, which could result in severe injury or death.

5.2.4.5 Inlet Design

Typically, standard culvert end sections are sufficient, however there are some situations where additional analysis is necessary. Locations where piping or buoyancy may occur, or where improved inlets are being considered will require additional evaluation and a special design.

The allowable headwater may not be the only design consideration especially when designing culverts in locations with high fill, steep fill slopes or that have a potential for high headwaters. It may be necessary to evaluate the potential for piping along the outside of the culvert. Piping can begin with seepage along the outside of the pipe that progressively removes the embankment material. The loss of embankment material can lead to pipe failure. If necessary, install mitigation to prevent piping. Piping mitigation measures include using joints that are as watertight as practical. Installing clay caps and culvert drop walls are also used to prevent seepage along culverts.

Anti-seepage collars are no longer widely used due to the potential that they impede compaction of soils around the pipe, which can lead to differential settlement, pipe deflection, or cracking of the pipe. Led by the dam safety community, anti-seepage measures such as filter diaphragms can be used to decrease the hydraulic gradient and decrease the velocity of flow along the outside of the pipe. A filter diaphragm is a zone of filter material constructed as a diaphragm surrounding a conduit through and embankment. The filter diaphragm is placed on all sides of the conduit and extends a distance into the embankment (FEMA 2011; ASDSO).

Inlet protection may be needed to prevent buoyancy that can lead to pipe failure. Flotation is used to describe the failure of a culvert due to the uplift pressures caused by buoyancy. Buoyancy typically occurs in a culvert in inlet control with a submerged upstream end when the pressure outside the culvert is less than the pressure in the barrel. It may also occur when the inlet is damaged or debris blocks the culvert end. Buoyancy is affected by steepness of the culvert slope, depth of the potential headwater, flatness of the upstream fill slope, height of the fill, larger culvert skews, or inlets with mitered ends.

Generally, only flexible pipes are vulnerable to this type of inlet failure. Inlet protection is usually necessary to anchor the inlet end of the culvert and provide buoyancy protection for some flexible culverts. The standard apron end section (Plate 3123) is considered adequate flotation protection for corrugated metal culverts 12” through 84”. However, flexible pipe with mitered end sections greater than 36” and all structural plate culverts should have buoyancy protection. Buoyancy mitigation measures provide anchorage of the culvert ends. Typical practices include slope paving around mitered metal end sections, installation of concrete or sheet pile cutoff walls, or headwalls.

Vortices, water flowing in a circular motion, can form at culvert inlets with submerged inlets. Vortices can reduce culvert capacity but the impacts are highly variable and difficult to predict. Vortex suppression in culvert design applications is not recommended due to concerns about them acting as siphons during high headwater, questions about dependability under most culvert flow conditions, and potential of creating a safety hazard. In locations where vortices cause erosion around the inlet, erosion protection measures such as riprap or other slope protection measures should be considered.

Improved inlets are an option for long culverts which will operate under inlet control. Culverts operating under inlet control generally flow partly full. In that situation it is possible to use a smaller pipe having about the same cross-sectional area as the area of flow by using special inlets such as the side-tapered or slope-tapered inlets. These inlets can be designed for both boxes and round pipe. While improved inlets can increase the hydraulic performance of the culvert, they will also add to the total culvert cost, will require a special structural design and fabrication, debris may be an issue, and may require additional maintenance. Therefore, these inlets should only be used when necessary. MnDOT’s standard flared inlets are not considered to be improved inlets. Additional information about tapered inlets is available in *Hydraulic Design of Highway Culverts* (FHWA 2012).

Another method to reduce culvert size is by utilizing a concrete pipe reducer section (Standard Plate 3001) to decrease the size of the culvert pipe needed after the water enters the pipe inlet. Reducers will require a through hydraulic analysis over a range of discharges and should only be used in locations where debris and the assumptions of inlet control will not be an issue. This method sizes the inlet by conventional means utilizing the inlet control nomographs. Then the size and location of the reducer is calculated in order to minimize the size of the culvert for the balance of the length. Two significant factors need to be considered: 1) the amount of reduction, and 2) the location of the reducer. The allowable diameter reduction is determined by comparing critical depth of the smaller pipe with its diameter. Too much constriction of the flow area can cause a choking effect, creating orifice flow at the reducer section throat, by sealing off the throat entrance to the smaller pipe with standing waves and other losses in the reducer. An orifice is an opening, submerged on the upstream side and flowing freely on the downstream side, which functions as a control section.

The longitudinal location of the reducer is based on the assumption that the velocity of flow entering the reducer should be equal or greater than the critical velocity in the smaller pipe. Starting with the critical depth of the smaller pipe, the standard incremental energy equation is used in determining the length of the larger pipe and its associated velocity. The velocity used to determine the total length of the pipe should be 10% greater than the critical velocity of the smaller pipe to help overcome any other losses that are developed by the flow contraction. Since the depth of flow crosses critical depth in the larger pipe at an approximate distance of 0.5 times the diameter from the inlet of the larger pipe, this length should be added to the total length of the pipe.

5.2.4.6 Outlet Protection

Outlet protection is needed when the velocities at the culvert outlet are high enough to cause scour or erosion. Protection against scour at culvert outlets ranges from riprap placement to complex and expensive energy dissipation devices. The most common energy dissipation methods used by MnDOT for roadside ditch installations are vegetative stabilization or riprap apron.

For outlet velocities less than or equal to 6 fps, check shear stress to determine if vegetation will be adequate (see Chapter 4). If vegetation is used, consider temporary erosion control during and immediately following construction until vegetation becomes established. If vegetation is not adequate, riprap aprons may be used. Riprap aprons for minor culverts (equivalent diameter ≤ 48”) are detailed in Std. Plates 3133 and 3134 and will generally be adequate. Use riprap apron recommendations provided in Table 5.5. Geotextile fabric may be substituted for granular filters. Additional information on riprap classes and material specifications are given in MnDOT *Standard Specifications for Construction*.

Table 5.5 Riprap Apron Recommendation Based on Outlet Velocity

Outlet velocity, V_o	**Riprap specifications**	**Filter specifications**
$0 < V_o \leq 6$ fps	Riprap or stable vegetation	varies
$6 < V_o \leq 8$ fps	12” Class II riprap	6” granular or geotextile filter
$8 < V_o \leq 10$ fps	18” Class III riprap	9” granular or geotextile filter
$10 < V_o \leq 12$ fps	24” Class IV riprap	12” granular or geotextile filter
$V_o > 12$ fps	Consider other energy dissipator	

If stabilized vegetation or riprap apron is ineffective, other alternatives may exist. These outlet protection alternatives are briefly described below. Additional details are also provided in Chapters 4 and 6.

Alternative energy dissipation considerations include, but are not limited to:

- A ring dissipator is used where right of way area is limited, debris is not a problem, the Froude Number (Fr) is >1.0, a riprap apron is not adequate, and moderate velocity reduction is needed. Reinforced concrete pipe energy dissipator options are located on Standard Plate 5010.
- A drop box inlet was developed by Soil Conservation Service (SCS) as a grade control structure. It can also be used at the culvert inlet to flatten the slope of the culvert and thereby reduce the velocity.
- A riprap basin may be used where adequate right of way is available, the Fr is less than 3.0, only a moderate amount of debris is present, adequate riprap is available, and other methods are not appropriate or are more expensive.
- An impact basin is used when little debris potential exists, design discharge is less than 150 cubic feet per second (cfs), no tailwater is required for successful operation, and is economically feasible.
- A stilling basin (SAF Basin) utilizes a hydraulic jump to dissipate energy. This option is an economical mean of dissipating large amounts of energy, but a tailwater is required and the Fr must be between 1.7 and 17.

5.2.5 Site Criteria

Design criteria depends on site factors such as structure type, length, location in plan, location in profile, overfill, debris and ice. The length of a culvert should be based on roadway clear zone and embankment geometry. Survey information used in culvert design includes topographic features, channel characteristics, fish migration needs, high water information if available, existing structures, and other related site-specific information.

5.2.5.1 Culvert Placement

Culvert location in both plan and profile should be investigated and designed to avoid sediment build-up in culvert barrels. In some situations, such as to improve AOP or floodplain connectivity, having the invert of one barrel lower than the others in multiple barrel crossings may be appropriate.

Locate and design culvert ends to create a reasonable roadside recovery area or clear zone for vehicles that may leave the roadway. This may require extending the pipe outlet outside the clear zone or using a traversable outlet such as a safety apron or safety apron and grate.

Consider channel alignment when determining the location and profile (slope) of the culvert.

- Severe or abrupt changes to channel alignment upstream or downstream of culverts are not recommended.
- Large culverts perpetuating drainage in defined channels should be skewed as necessary to minimize channel relocation and erosion.

- At most locations, the culvert profile should approximate the natural stream profile. Exceptions can be considered to arrest stream degradation by utilizing a drop inlet, broken-back culvert, or improve the hydraulic performance by utilizing a slope tapered inlet.
- Small culverts with no defined channel are placed normal to roadway centerline.

5.2.5.2 Minimum and Maximum Overfill

Minimum cover and maximum fill heights for Reinforced Concrete Pipe (RCP), Corrugated Steel Pipe (CSP), and plastic pipe are provided in Chapter 2.

All box culverts and bridges require that a structural engineer design and prepared plan. The Bridge Standard Plans Manual is available for box culvert design. Precast box culverts with fill heights less than 2.0 feet require a distribution slab.

Culverts located under deep fill are difficult to replace. In critical locations, upsizing a new culvert to allow for a future lining or other trenchless construction may be considered.

5.2.5.3 Debris and Ice

The potential for plugging with debris or ice should be assessed. The source of the debris or ice and the potential flood damage resulting from plugged culverts are important. Options available to the designer include:

- Attempt to pass the debris or ice through the culvert, which is usually accomplished by the placement of relief openings (preferred alternative) or increasing the culvert height.
- Retain the debris or ice upstream of the culvert (may require more frequent maintenance).
- Use non-flared sloped end sections that allow the ice and debris to ride up the sloped end.
- Use a bridge.

Where experience or physical evidence indicates the watercourse will transport a heavy volume of debris, the following information should be considered prior to a decision of whether to attempt to pass or retain the debris:

- the type and quantity of debris
- experience of upstream or downstream culverts in passing or retaining the debris
- experience of the subject roadway passing debris at the sag point
- whether large floatable debris will usually ride up the culvert end sections
- availability of access for maintenance to remove debris from the culvert entrance or the debris barrier
- assessment of damage due to debris clogging, if protection is not provided
- feasibility of relief opening, either in the form of a vertical riser or a relief culvert placed higher in the embankment
- guidance of HEC-9, *Debris Control Structures* (FHWA 2005)

5.2.5.4 Multiple Use Culverts

Consideration may be given to combining drainage culverts with other land use requirements such as providing passage under a highway for animals, boat traffic, or pedestrians. During the selected design discharge the full culvert may convey water and thus cannot be used for passage under the highway. For dual use culverts multiple barrels are preferred with at least one barrel available to convey runoff from more frequent flood events so that land use passage is still available.

Design considerations include:

- Two or more barrels are required with one situated to be dry during floods less than the selected design discharge.
- The culvert should be sized to ensure it can serve its intended land use function up to and including at a minimum the 2-year design event.
- The height and width constraints are designed to satisfy the hydraulic or land use requirements, whichever is larger.

5.2.6 Aquatic Organism Passage (AOP)

Culvert crossings in areas with a high likelihood of aquatic organism use should also be designed using the principles of aquatic organism passage. Best practices for achieving stream connectivity and AOP include:

- Design culvert slope to match stream channel slope.
- Align the culvert to match the adjacent channel alignment.
- Set the culvert opening width to bankfull channel width or slightly greater.
- Design for comparable flow depth and velocity within the culvert to adjacent channel.
- Provide a continuous sediment bed with similar roughness to adjacent channel.
- Maintain continuity of sediment transport and debris passage through culvert similar to adjacent channel reaches.
- Design for safety to the traveling public, longevity, and resilience.

Successful AOP designs incorporate as many best practices as practicable. For further information see *Minnesota Guide for Stream Connectivity and Aquatic Organism Passage Through Culverts* (MnDOT 2019).

5.2.7 Design Methods and Computer Applications

The designer has several computational methods to choose from when designing a culvert. These methods include: nomographs, hand calculations, or computer applications. Computer applications are commonly used and have the advantage of being able to rapidly perform iterations in order to compare different designs. Computer applications solve equations and give answers, but it does not mean the answers are correct. The designer still needs to understand culvert design and be able to assess the reasonableness of the solution. Occasional hand or nomograph computations are a good way to perform a quick check of computer results. Common applications to design or analyze culverts include HY-8 and HEC-RAS. The Nebraska DOT Broken-back Culvert Analysis Program (BCAP) is one of the few programs that can analyze a hydraulic jump. BCAP may be necessary for design of culverts on a long, steep slope or where a hydraulic jump is desired to reduce the outlet velocity.

Culverts are designed by assuming either a constant discharge (peak flow) or routing a hydrograph. Constant discharge is utilized for most culvert designs. The analysis is performed using the peak discharge; however, a range of discharges and a performance curve is recommended for the major culverts. Using a constant discharge will yield a conservatively sized structure where temporary storage is available but is not considered in the design. Acceptable methods of determining the peak discharge are detailed in Chapter 3 and include Regression Equations developed by United States Geological Survey (USGS) for ungaged streams, Log Pearson III analysis for gaged streams, SCS Method, and Rational Method (for drainage areas less than 200 acres).

Flood routing through a culvert is a practice that evaluates the effect of temporary upstream ponding caused by the culvert's backwater. Flood routing requires synthesis of a hydrograph and should be used when significant storage upstream will reduce the required culvert size, or storage capacity behind a highway embankment attenuates a flood hydrograph and reduces the peak discharge. Guidance on design considerations is listed in Section 5.2.3 under allowable headwater and upstream storage. Flood routing equations and procedures are detailed in Chapter 7. In some complex situations where multiple hydraulic infrastructure interact, or water elevations get high enough to cross a drainage divide or embankment, additional analysis may be needed to accurately model the flow patterns, flow rates and water surface elevations.

5.3 Culvert Analysis

The procedures in this chapter utilize the concepts of minimum performance and control sections. The concept of minimum performance means that although the culvert may operate more efficiently at times, (more flow for a given headwater level), it will never operate at a lower level of performance than calculated. Minimum performance is assumed by analyzing both inlet and outlet control and using the highest headwater.

The control section is the location where there is a unique relationship between the flow rate and the upstream depth of water. Inlet control is governed by the inlet geometry which includes the barrel shape, cross-sectional area, and inlet edge. Outlet control is governed by a combination of the culvert inlet geometry, the barrel characteristics, and the tailwater.

An exact analysis of culvert flow is complex because of the following requirements:

- Analysis of nonuniform flow with regions of both gradually varying and rapidly varying flow
- Determination of how the flow type changes as the flow rate and tailwater elevations changes
- Application of backwater and drawdown calculations, energy and momentum balance

- Application of hydraulic model results
- Determination of the location of any hydraulic jumps that may occur, and whether inside or downstream of the culvert barrel

5.3.1 Inlet Control

For inlet control, the control section is at the upstream end of the barrel (the inlet). The flow passes through critical depth near the inlet and becomes shallow, high velocity (supercritical) flow in the culvert barrel. Depending on the tailwater, a hydraulic jump may occur downstream of the inlet, either inside or outside of the pipe. The relationship between flow and water surface elevation must be determined by model tests of the weir geometry or by measuring prototype discharges. These tests are then used to develop equations. Appendix A of HDS-5 (FHWA 2012) contains these equations, which were developed from model test data.

Factors that influence inlet control include headwater, inlet area, inlet shape, inlet configuration, and barrel shape. Headwater depth is measured from the inlet invert to the surface of the upstream pool. Inlet area is the cross-sectional area of the face of the culvert. Often the inlet area is the same as the barrel area. Inlet configuration describes the entrance type which includes apron inlet, beveled edge, mitered to conform to slope, socket end, square edge in a headwall, and thin edge projecting. Inlet shape is usually the same as the shape of the culvert barrel. Typical shapes are rectangular, circular, elliptical, and arch. Check for an additional control section such as an improved inlet where the geometry varies.

Three regions of flow are shown in Figure 5.5, unsubmerged (weir flow), transition and submerged (orifice flow) zones. An unsubmerged inlet occurs when the headwater is below the inlet crown ($HW<D$) and the entrance operates as a weir. The transition zone is located between the unsubmerged and the submerged flow conditions where the flow is poorly defined. This zone is approximated by plotting the unsubmerged and submerged flow equations and connecting them with a line tangent to both curves.

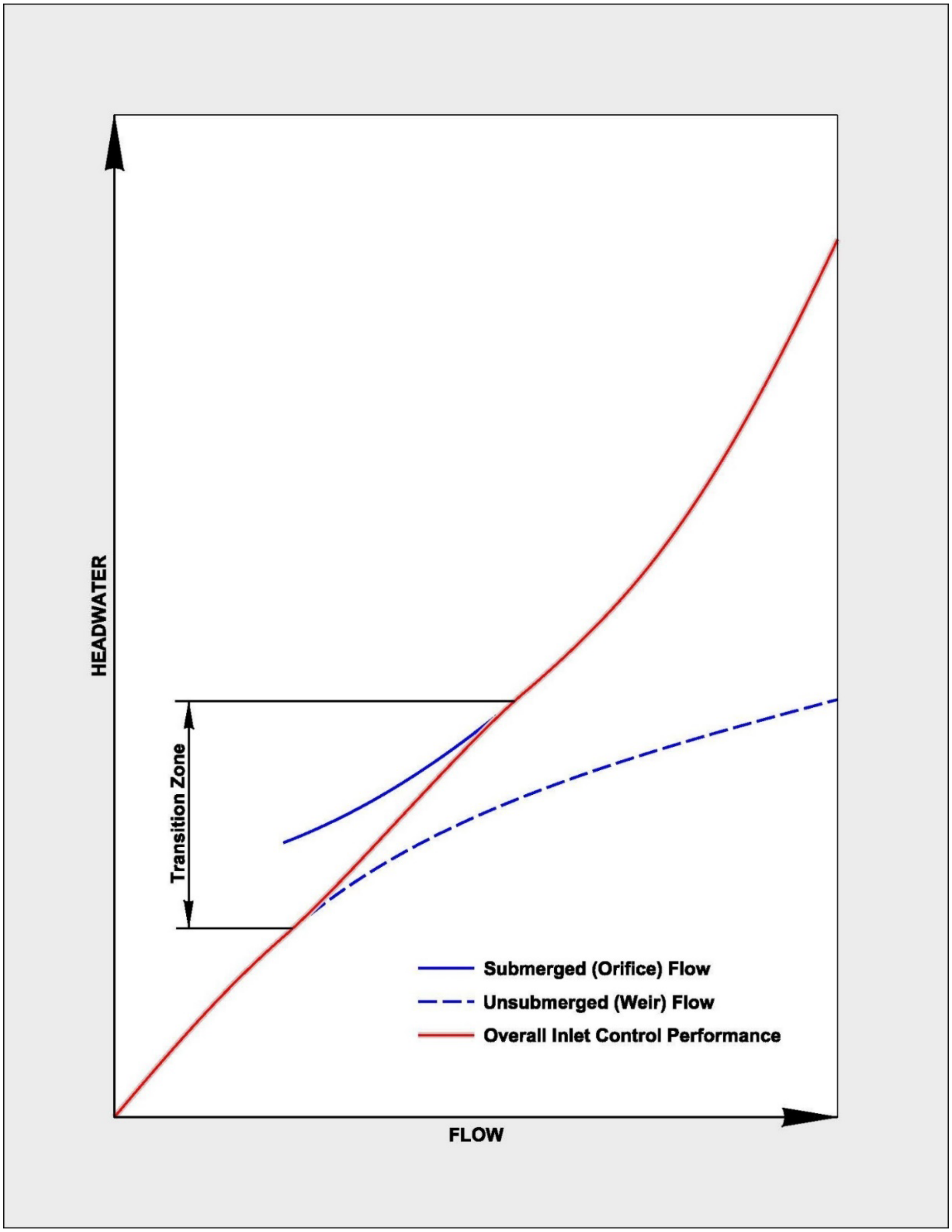


Figure 5.5 Inlet Control Curves
Source: *HDS-5, Figure 3.4* (FHWA 2012)

A submerged inlet occurs when the headwater is above the inlet and the culvert inlet operates as an orifice. The relationship between flow and headwater can be defined based on results from model tests. Appendix A of *HDS-5* (FHWA 2012) contains flow equations which were developed from model test data. Figures 5.6 and 5.7 show submerged inlets under inlet control, one with the tailwater higher than critical depth and the other with the tailwater at or lower than critical depth.

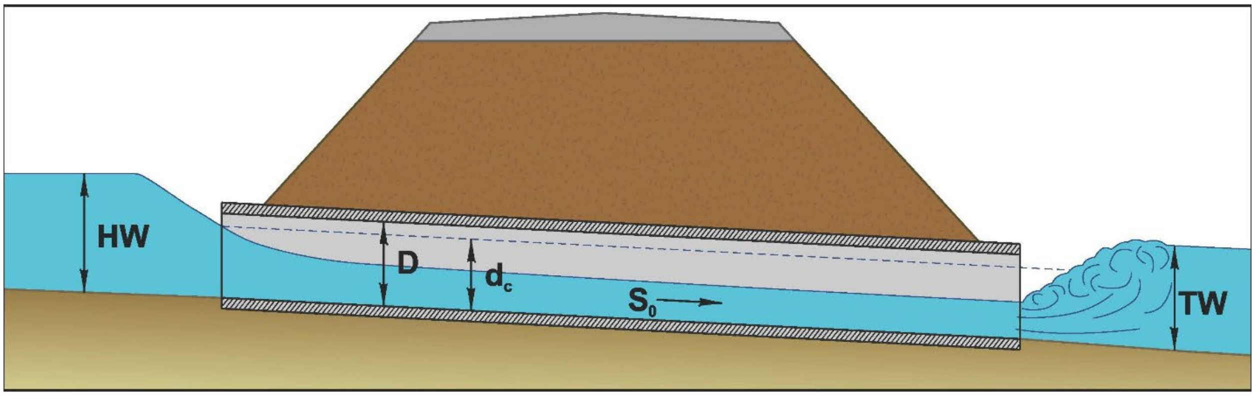


Figure 5.6 Submerged Inlet in Inlet Control with Tailwater Higher than Critical Depth
Source: *HDS-5, Figure 3.25* (FHWA 2012)

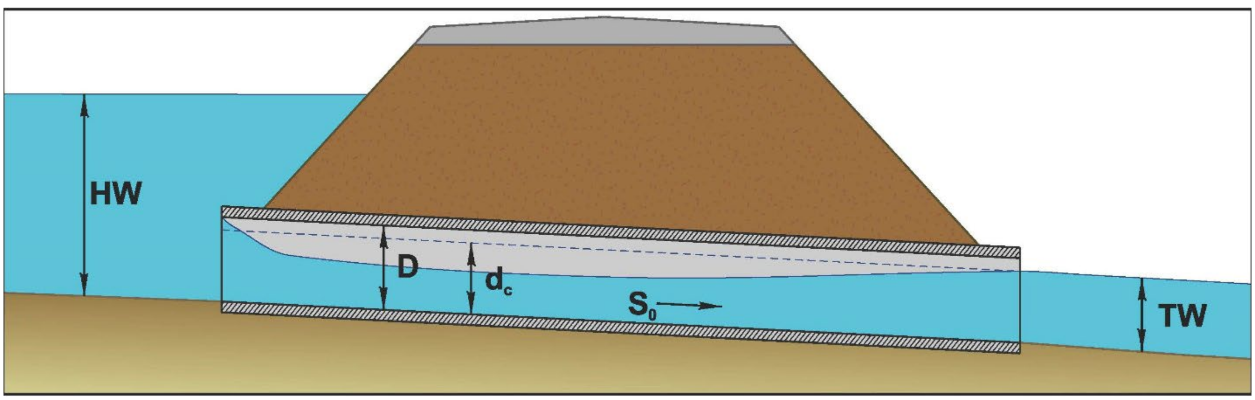


Figure 5.7 Submerged Inlet in Inlet Control with Tailwater at or Lower than Critical Depth
Source: *HDS-5, Figure 3.26* (FHWA 2012)

The inlet control flow versus headwater curves which are established using the above procedure are the basis for constructing the inlet control design nomographs. Note that in the inlet control nomographs (Appendix F), headwater (HW) is measured to the total upstream energy grade line including the approach velocity head. Most designers now use computer applications to compute the inlet headwater during inlet control. Not all computer programs use the same assumptions and results may vary. It may be necessary to check computer application results against nomographs. Culvert design nomographs are available in Appendix F.

5.3.2 Outlet Control

Outlet control has depths and velocity which are subcritical. The control of the flow is at the downstream end of the culvert (the outlet) or further downstream in the channel. Tailwater (TW) is based on the downstream water surface elevation and is assumed to be critical depth near the culvert outlet or the downstream channel depth, whichever is higher. Backwater calculations from a downstream control, a normal depth approximation, or field observations are used to define the tailwater elevation. In a given culvert, the type of flow is dependent on all the barrel factors: barrel roughness, barrel area, barrel length and barrel slope. The inlet control factors also influence culverts in outlet control.

In the analysis of outlet control hydraulics, full flow condition is assumed when TW depth is above the crown of the culvert. The outlet control nomographs are accurate for the analysis of full flow condition. Partial full condition is used when TW depth is below the crown of the culvert. A backwater calculation is necessary to accurately analyze partial flow conditions, however an approximate method utilizing TW = $(d_c+D)/2$ yields reasonable results, especially with culverts less than 48” in diameter. Outlet control flow conditions can be calculated based on an energy balance from the tailwater pool to the headwater pool.

Velocity is computed by rearranging the continuity equation. Keep in mind that unless a round or arch pipe is full, the formula for determining area used in this equation is complex. The alternative is to use the culvert design nomographs located in Appendix F.

$$V = \frac{Q}{A}$$

Equation 5.2

Where:

V = velocity (ft/sec)
Q = flow rate (cfs)
A = cross-sectional area of the flow area in the barrel (ft^2)

Head Losses

Full flow in the culvert barrel, as depicted in Figure 5.8, is the best flow type for describing the hand computation of outlet control hydraulics. Outlet control flow conditions can be calculated based on an energy balance from the tailwater pool to the headwater pool. The total energy (H_L) required to pass the flow through the culvert barrel is made up of the entrance loss (H_e), the friction losses through the barrel (H_f), and the exit loss (H_o). Other losses, including bend losses (H_b), losses at junctions (H_j), and losses at grates (H_g) should be included as appropriate. These other losses occur less frequently for most culverts, use of these losses is discussed in Chapters 2 and 5 of *HDS-5* (FHWA 2012).

$$H_L = H_e + H_f + H_o + H_b + H_j + H_g$$ **Equation 5.3a**

Where:

g = acceleration due to gravity (32.2 ft/s^2)
H = barrel losses (ft)
H_b = bend losses (ft)
H_e = entrance loss (ft)
H_f = friction losses (ft)
H_g = losses at grates (ft)
H_j = junction losses (ft)
H_L = total energy head loss (ft)
H_o = exit loss (ft)
H_v = full flow velocity head (ft)
k_e = entrance loss coefficient (Tables 5.4 and 5.6)
L = length of the culvert barrel (ft)
n = Manning's roughness coefficient (Table 5.7)
P = wetted perimeter of the barrel (ft)
R = hydraulic radius of the full culvert barrel (ft) = area/wetted parameter
V = average barrel velocity (ft/s)
V_d = channel velocity downstream of the culvert (ft/s)

Entrance Loss

$$H_e = k_e\left(\frac{V^2}{2g}\right)$$ **Equation 5.3b**

Friction Loss

$$H_f = \left(\frac{29n^2L}{R^{1.33}}\right)\left(\frac{V^2}{2g}\right)$$ **Equation 5.3c**

Velocity Head

$$H_v = \frac{V^2}{2g}$$ **Equation 5.3d**

Exit Loss

$$H_o = 1.0\left[\left(\frac{V^2}{2g}\right) - \left(\frac{V_d^2}{2g}\right)\right]$$ **Equation 5.3e**

V_d is usually neglected, then:

$$H_o = H_v = \frac{V^2}{2g}$$ **Equation 5.3f**

Barrel Losses

$$H = H_e + H_o + H_f$$ **Equation 5.3g**

Substituting in equations:

$$H = \left[1 + k_e + \left(\frac{29n^2L}{R^{1.33}}\right)\right]\left[\frac{V^2}{2g}\right]$$ **Equation 5.4**

Entrance loss coefficients for end sections included in the MnDOT Standard Plates are provided in Table 5.4. Table 5.6 has entrance loss coefficients for additional end section options. Concrete and metal end sections conforming to fill slopes refers to the sections commonly available from manufacturers. From limited hydraulic tests, these end sections are equivalent in operation to a headwall under either inlet or outlet control. Some end sections, which incorporate a closed taper in their design, have superior hydraulic performance. These latter sections can be hydraulicly designed using the information given for the beveled inlet but are not available in MnDOT Standard plates so will require a structural design and design detail in the plan.

Table 5.6 Entrance Loss Coefficients, k_e for Outlet Control, Full or Partly Full

Type of structure	Design of Entrance	Coefficient k_e
Concrete pipe	Mitered to conform to fill slope	0.7
Concrete pipe	End-section conforming to fill slope	0.5
Concrete pipe	Projecting from fill, square cut end	0.5
Concrete pipe	Headwall or headwall and wingwall: Square-edge	0.5
Concrete pipe	Headwall or headwall and wingwall: Rounded (radius=1/12D)	0.2
Concrete pipe	Headwall or headwall and wingwall: Socket end of pipe (grooved-end)	0.2
Concrete pipe	Projecting from fill, socket end (grooved-end)	0.2
Concrete pipe	Beveled edges, 33.7° or 45° bevels	0.2
Concrete pipe	Side or slope tapered inlet	0.2
Corrugated metal pipe	Projecting from fill (no headwall)	0.9

Type of structure	Design of Entrance	Coefficient k_e
Corrugated metal pipe	Mitered to conform to fill slope, paved or unpaved slope	0.7
Corrugated metal pipe	Headwall or headwall and wingwalls square-edge	0.5
Corrugated metal pipe	End-section conforming to fill slope	0.5
Corrugated metal pipe	Beveled edges, 33.7° or 45° bevels	0.2
Corrugated metal pipe	Side or slope tapered inlet	0.2
Reinforced concrete box	Wingwalls parallel (extension of sides), square-edged at crown	0.7
Reinforced concrete box	Wingwalls at 10°to 25° to barrel, square edged at crown	0.5
Reinforced concrete box	Headwall parallel to embankment (no wingwalls): Squared edged on 3 edges	0.5
Reinforced concrete box	Headwall parallel to embankment (no wingwalls): Rounded on 3 edges to radius of 1/12 barrel dimension	0.2
Reinforced concrete box	Headwall parallel to embankment (no wingwalls): Beveled edges on 3 sides	0.2
Reinforced concrete box	Wingwalls at 30° to 75° to barrel: Crown edge rounded to radius of 1/12 barrel dimension	0.2
Reinforced concrete box	Wingwalls at 30° to 75° to barrel: Beveled top edge	0.2
Reinforced concrete box	Wingwalls at 30° to 75° to barrel: Square edge crown	0.4
Reinforced concrete box	Side or slope tapered inlet	0.2

Notes:

1. Source: HDS-5 (FHWA 2012)
2. “End section conforming to fill slope” refers to metal or concrete pipe flared apron end-sections commonly available from manufacturers.

Design Manning’s n values are provided in Table 5.7. These are the numbers that should be used for the design of new culverts to provide consistency in the hydraulic analysis. When multiple plastic pipe alternatives are allowed it is easier to analyze if they have the same Manning’s n. For generic pipe where multiple pipe materials are allowed it will be necessary to use the most conservative Manning’s n for design since the designer will not know which pipe material will be installed. This means the material with the highest Manning’s n should be used to design the pipe size, otherwise the pipe capacity and headwater could be underestimated. Conversely, for outlet velocity and design of outlet protection, using the material with the lowest Manning’s n will ensure the pipe outlet is stable.

Table 5.7 Manning's n Values for Culverts

Type of culvert	Roughness or Corrugation	Laboratory Manning's n	Design Manning's n
Reinforced concrete pipe	Smooth	0.010 - 0.011	0.012
Concrete box	Smooth	0.012 - 0.015	0.013
Spiral rib metal pipe	Smooth	0.012 - 0.013	0.013
Corrugated metal pipe (CMP)	2 by ½ inch annular	0.022 - 0.027	0.024
Corrugated metal pipe (CMP)	2 by ½ inch helical	0.011 - 0.023	0.024
Corrugated metal pipe (CMP)	6 by 1 inch	0.022 - 0.025	0.027
Corrugated metal pipe (CMP)	5 by 1 inch	0.025 - 0.026	0.027
Corrugated metal pipe (CMP)	3 by 1 inch	0.027 - 0.028	0.027
Corrugated metal pipe (CMP)	6 by 2 inch structural plate	0.033 - 0.035	0.036
Corrugated metal pipe (CMP)	9 by 2½ inch structural plate	0.033 - 0.037	0.036
Corrugated polyethylene pipe, dual wall (CP)	Smooth liner	0.009 - 0.015	0.012
Corrugated polypropylene pipe, dual wall (PP)	Smooth liner	Not Listed	0.012
Polyvinyl chloride pipe (PVC)	Smooth	0.009 - 0.011	0.012

Notes:
1. Source: excerpt from Table B.1 HDS-5 (FHWA 2012)
2. Laboratory Manning's n values were obtained in a controlled environment.
3. Actual field values for culverts may vary depending on the effect of abrasion, corrosion, deflection, deterioration, and joint conditions.
4. Use MnDOT recommended Design Manning's n values when designing new culverts.
5. Manning's n values for CMP varies for different corrugation dimensions, barrel size, and between annular and helical pipe.

Stage Increase

The stage increase (S.I.) is the increase in water surface elevation due to the constriction of the culvert. It is computed as the difference between the water surface elevation prior to installation and the head water just upstream of the culvert. Headwater must be measured sufficiently distant from the pipe to exclude local drawdown effects.

$$S.I. = HW - (TW + LS_o)$$ **Equation 5.5**

Energy Grade Line

The energy grade line represents the total energy at any point along the culvert barrel. Equating the total energy at sections 1 and 2, upstream and downstream of the culvert barrel in Figure 5.8, the following relationship results:

$$HW_o + \frac{V_u^2}{2g} = TW + \frac{V_d^2}{2g} + H_L$$ **Equation 5.6**

Where:

- HW_o = headwater depth above the outlet invert (ft)
- V_u = approach velocity (ft/s)
- TW = tailwater depth above the outlet invert (ft)
- V_d = downstream velocity (ft/s)
- H_L = sum of all losses (Equation 5.2a)
- L = Length of culvert barrel (ft)
- S_o = slope of culvert invert (ft/ft)

Hydraulic Grade Line

The hydraulic grade line is the depth to which water would rise in vertical tubes connected to the sides of the culvert barrel. In full flow, the energy grade line and the hydraulic grade line are parallel lines separated by the velocity head except at the inlet and the outlet.

Full Flow

Nomographs were developed assuming that the culvert barrel is flowing full. HY-8 can duplicate the nomograph solution by assuming full flow in the barrel. Two flow conditions occur in outlet control when the barrel is flowing full: TW > D, Flow Type 4 (see Figure 5.8) and TW > d_c, Flow Type 6 (see Figure 5.9).

V_u is small and its velocity head can be considered to be a part of the available headwater (HW) used to convey the flow through the culvert. V_d is small and its velocity head can be neglected. Equation 5.6 becomes:

$$HW = TW + H - S_oL$$ **Equation 5.7**

Where:

- HW = depth from the inlet invert to the energy grade line (ft)
- H = the value read from the nomographs (ft) or Equation 5.4
- S_oL = drop from inlet to outlet invert (ft)

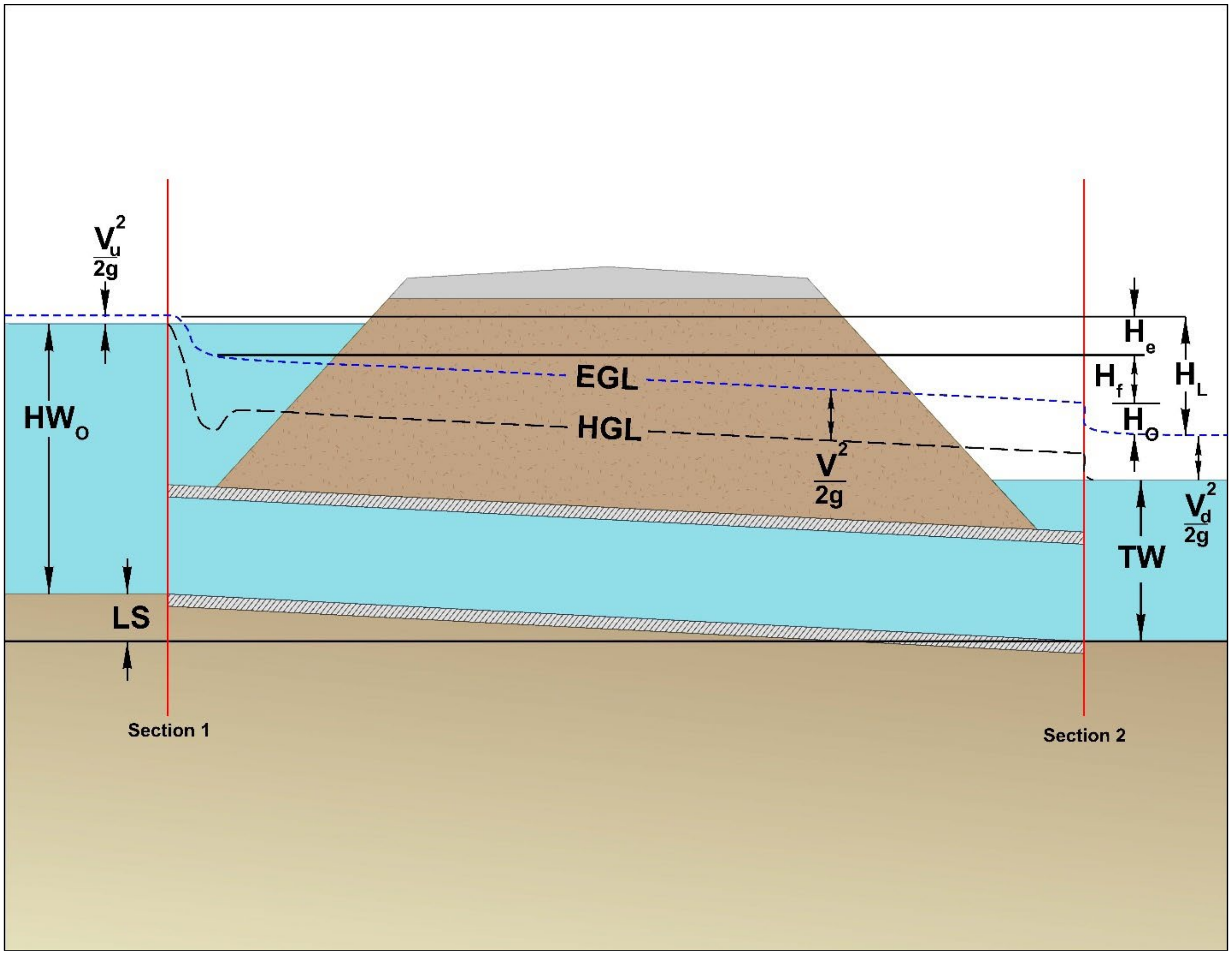


Figure 5.8 Full Flow Energy and Hydraulic Grade Lines
Source: *HDS-5, Figure 3.8* (FHWA 2012)

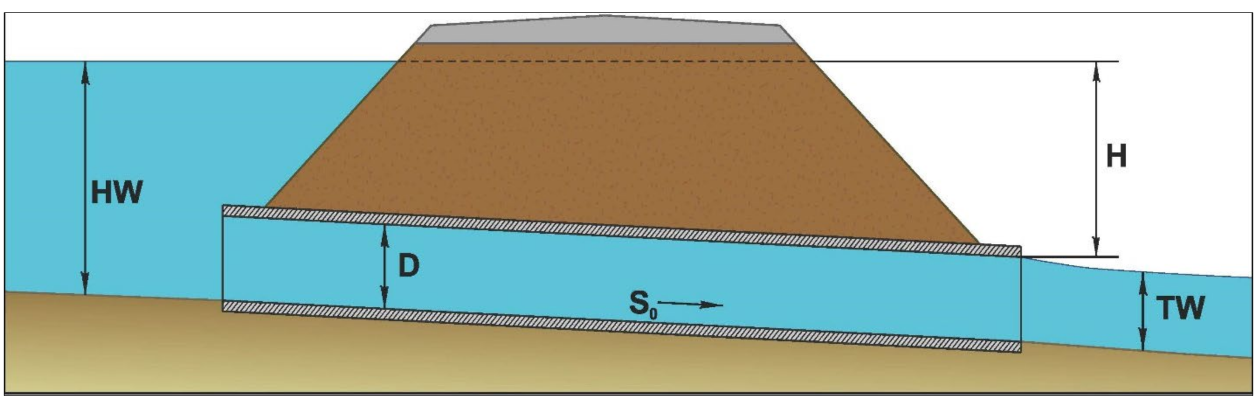


Figure 5.9 Flowing Full in Outlet Control, Where Tailwater is Higher than Critical Depth
Source: *HDS-5, Figure 3.29* (FHWA 2012)

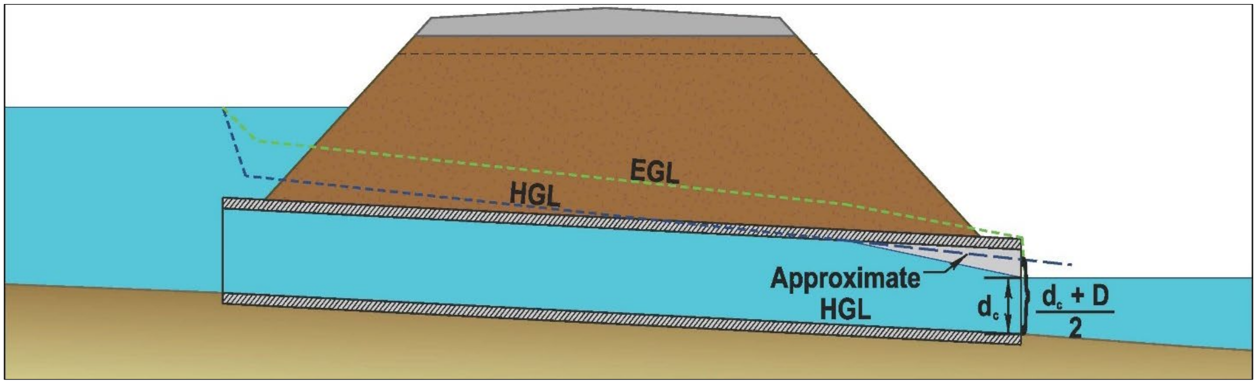


Figure 5.10 Effectively Full Flow in Outlet Control, Where Tailwater is At or Lower than Critical Depth
Source: *HDS-5, Figure 3.30* (FHWA 2012)

Partly Full Flow

Equations 5.2 through 5.6 were developed for full barrel flow. The equations also apply to the flow situations which are effectively full flow conditions, when TW < d_c, (see Figure 5.10).

When the culvert barrel is partly full, backwater calculations may be required which begin at the downstream water surface and proceed upstream. If the depth intersects the top of the barrel, full flow extends from that point upstream to the culvert entrance.

Based on model studies and numerous backwater calculations performed by FHWA, it was found that the hydraulic grade line pierces the plane of the culvert outlet at a point half-way between critical depth and the top of the barrel or (d_c + D)/2 above the outlet invert. TW should be used if higher than (d_c + D)/2. The following equation should be used for culverts that are partially full when determining the Headwater (HW) for outlet control:

$$HW = h_o + H - S_oL$$ **Equation 5.8**

Where:

h_o = the larger of TW or (d_c+D)/2 (ft)

Adequate results are obtained down to a HW = 0.75D. For lower headwaters and major culverts, backwater calculations are recommended. Figures 5.11 and 5.12 show partially full flow conditions under outlet control with TW < d_c and TW > d_c respectively.

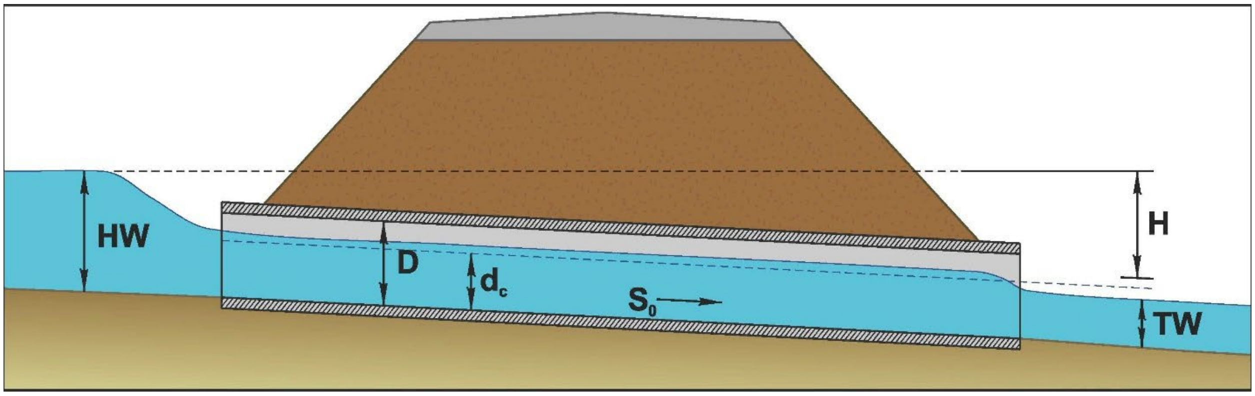


Figure 5.11 Partly Full Flow in Outlet Control, Where Tailwater is Lower than Critical Depth
Source: *HDS-5, Figure 3.27* (FHWA 2012)

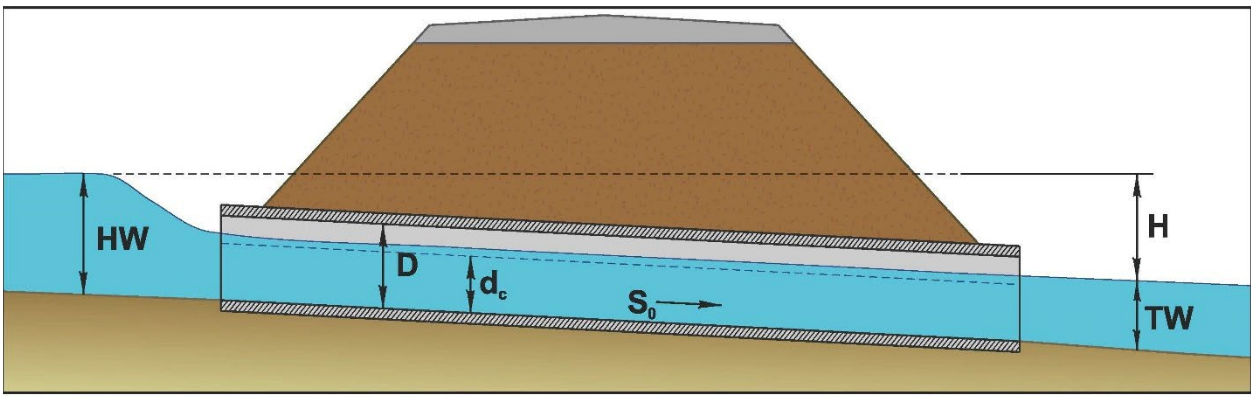


Figure 5.12 Partly Full Flow in Outlet Control, Where Tailwater is Higher than Critical Depth
Source: *HDS-5, Figure 3.28* (FHWA 2012)

5.3.3 Outlet Velocity

Culvert outlet velocities shall be calculated to determine the need for erosion protection at the culvert exit. Since the culvert outlet velocity is usually higher than the natural stream velocity, energy dissipation may be necessary to prevent downstream erosion. If outlet erosion protection is necessary, the flow depths and Froude number may also be needed for energy dissipation design. Outlet velocities for inlet and outlet control are computed as follows:

Inlet Control

If water surface profile calculations are necessary, begin at d_c at the entrance and proceed downstream to the exit. Determine depth and flow area at the exit. The velocity is calculated from Equation 5.2. While water surface profiles can be computed by hand, typically when water surface profile calculations are necessary a computer application will be selected to perform the iterative computations.

Use normal depth and velocity. This approximation may be used since the water surface profile converges towards normal depth if the culvert is of adequate length. The normal depth velocity may be higher than the actual velocity at the outlet determined from running a water surface profile. Normal depths and velocities may be obtained from the open channel flow charts for circular pipe provided in Appendix F.

Outlet Control

The cross-sectional area of the flow is defined by the geometry of the outlet and either critical depth, tailwater depth, or the height of the conduit. There are three types of outlet control:

- Critical depth is used when the tailwater is less than critical depth.
- Tailwater depth is used when tailwater is greater than critical depth, but below the top of the barrel.
- The total barrel area is used when the tailwater exceeds the top of the barrel.

5.3.4 Roadway Overtopping

Roadway overtopping will begin when the headwater rises to the elevation of the roadway and starts to flow over the road. Water may be on the road surface prior to overtopping the road and may result in the need to close the road prior to an overtopping occurrence. Culverts are designed for an allowable headwater that includes a safety factor that will reduce the risk of roadway flooding during the design event. Overtopping will usually occur at the low point of a sag vertical curve on the roadway, which may not be located near the culvert. The flow will be similar to flow over a broad crested weir.

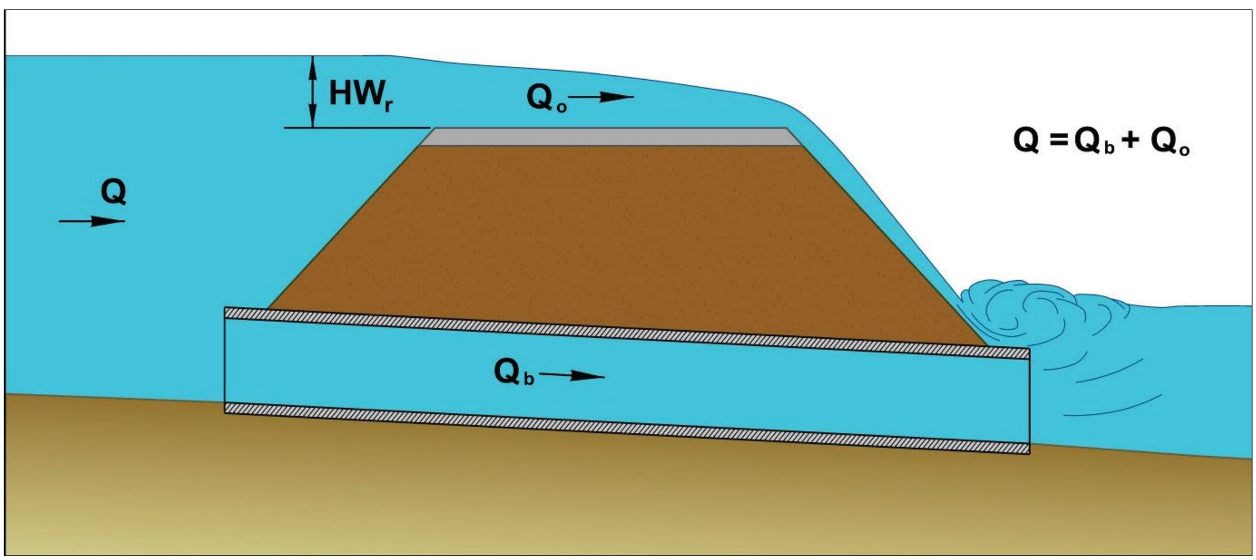


Figure 5.13 Roadway Overtopping
Source: *HDS-5, Figure 3.10* (FHWA 2012)

Equation 5.9a defines the flow across the roadway.

$$Q_o = C_d L (HW_r)^{1.5}$$ **Equation 5.9a**

$$Q = Q_b + Q_o$$ **Equation 5.9b**

Where:

Q_o = overtopping flow rate (cfs)
Q_b = culvert barrel flow rate (cfs)
Q = Total flow over the road and through the culvert (cfs)
C_d = overtopping discharge coefficient = k_t C_r
k_t = submergence coefficient
C_r = discharge coefficient
L = length of the roadway crest (ft)
HW_r = the upstream depth, measured above the roadway crest to water surface upstream of weir drawdown (ft)

Flow coefficients for flow overtopping roadway embankments are shown in Figure 5.14. Figure 5.14A is used for deep overtopping, Figure 5.14B is used for shallow overtopping and Figures 5.14C is a correction factor for downstream submergence which occurs as the tailwater begins to encroach on the free overfall from the weir.

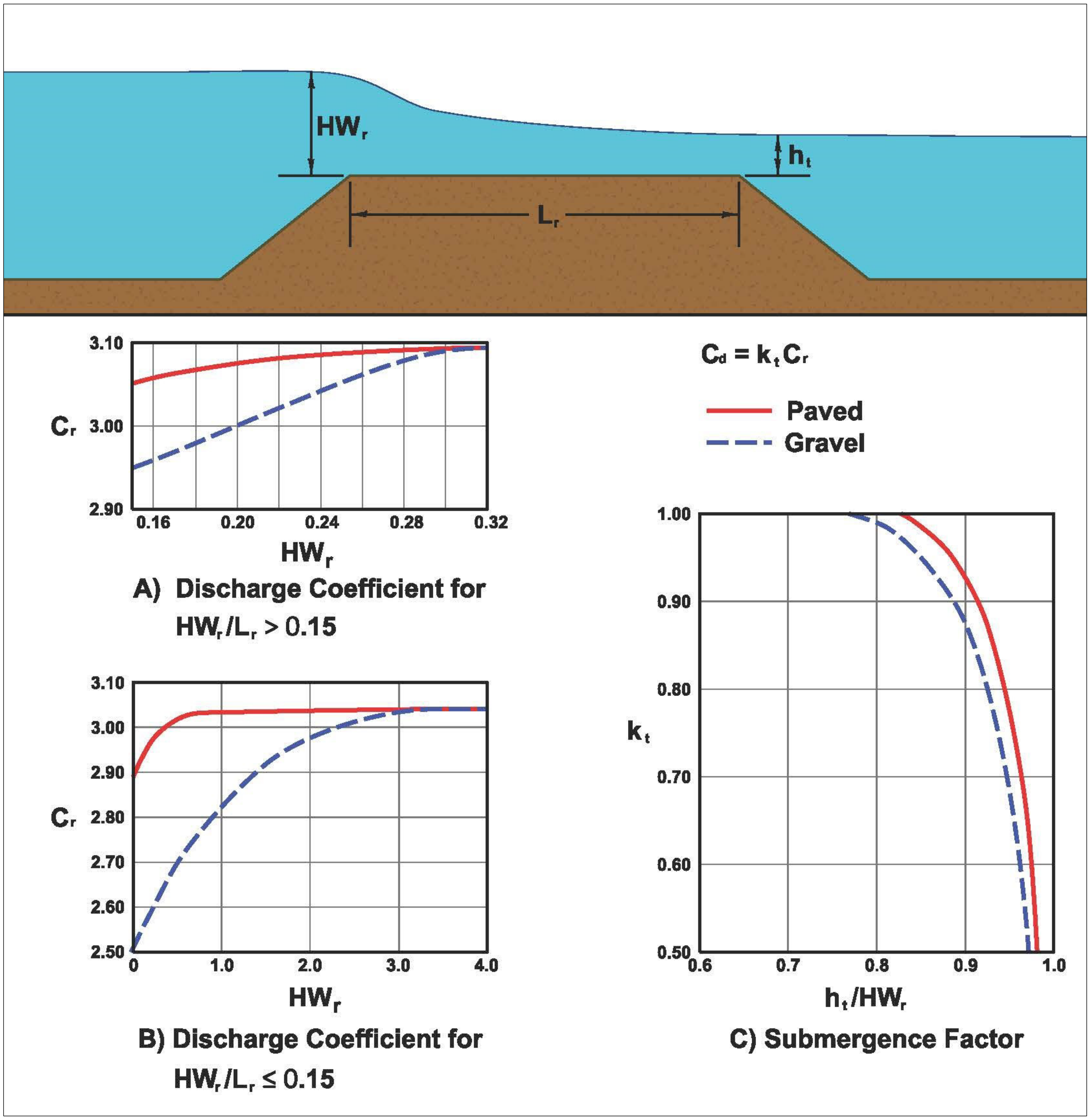


Figure 5.14 Discharge Coefficients for Roadway Overtopping
Source: *HDS-5, Figure 3.11* (FHWA 2012)

When the road is acting as a weir crest the length can be represented by a single horizontal line (one segment). The length of the weir is the horizontal length of this segment. The depth is the average depth of the upstream pool above the roadway. The length is difficult to determine when the crest is defined by a roadway sag vertical curve. In that situation the length should be subdivided into a series of segments. The flow over each segment is calculated for a given headwater. The flows for each segment are added together to determine the total flow. These two situations are illustrated in Figure 5.15.

Total flow at the culvert is calculated for a given upstream water surface elevation by adding the roadway overflow plus culvert flow. Performance curves for the culvert and the road overflow may be summed to yield an overall performance curve. A trial-and-error process is necessary to determine the flow passing through the culvert and the amount flowing across the roadway. Assume HW_r and solve for Q_o, then solve for Q_b through the culvert barrel by computing HW-TW and solving for V_b and then Q_b. Summation of the Q's must balance with the total Q.

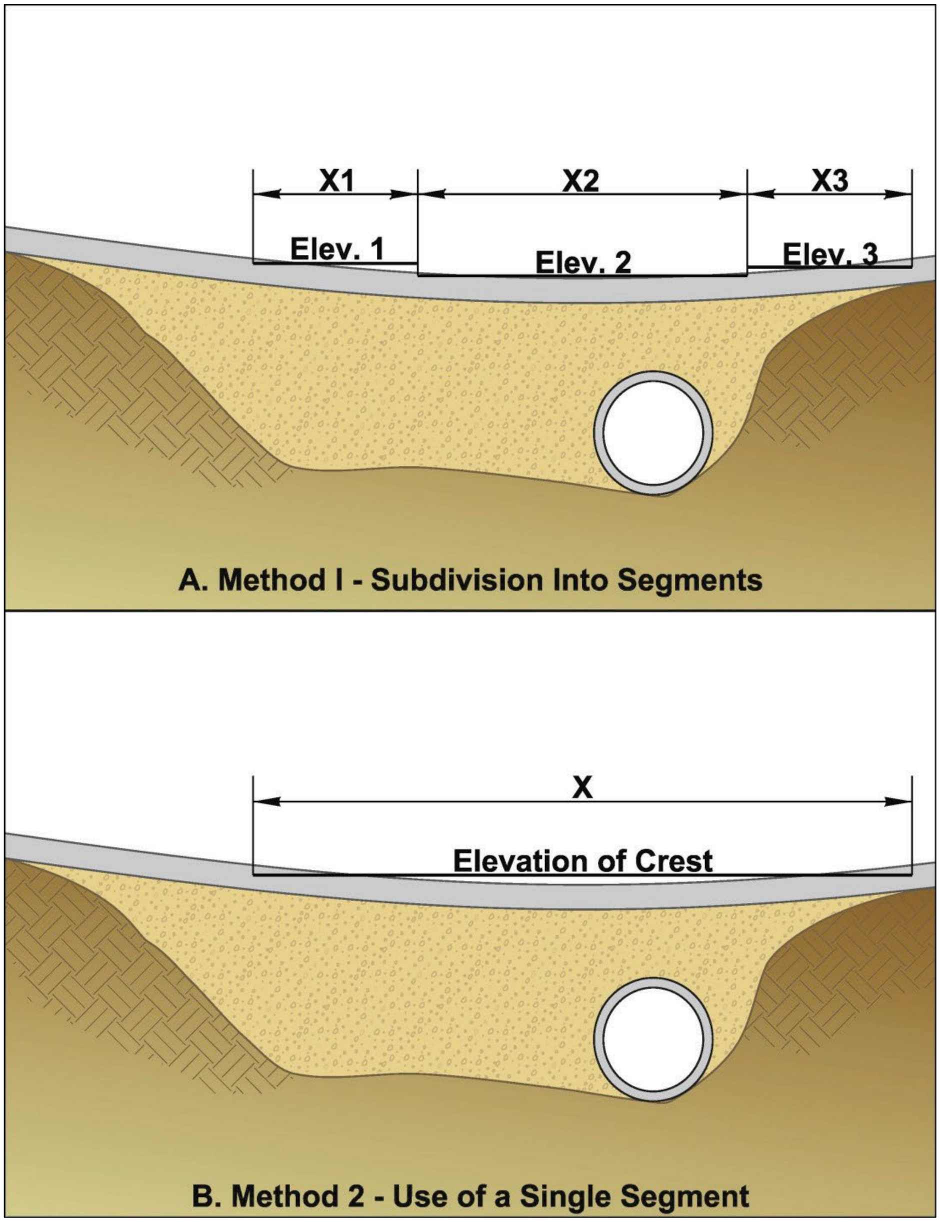


Figure 5.15 Weir Crest Length Determinations for Roadway Overtopping
Source: *HDS-5, Figure 3.12* (FHWA 2012)

5.3.5 Performance Curves

A performance curve is a plot of flow rate versus headwater depth or elevation. The culvert performance curve is made up of the controlling portions of the inlet, outlet, and roadway overtopping performance curves (See Figure 5.16). The overall performance curve is the sum of the flow through the culvert and the flow across the roadway and can be determined by performing the following steps.

Step 1 Select a range of flow rates and determine the corresponding headwater elevations for the culvert flow alone. These flow rates should fall above and below the design discharge and cover the entire flow range of interest. Both inlet and outlet control headwaters should be calculated.

Step 2 Combine the inlet and outlet control performance curves to define a single performance curve for the culvert.

Step 3 When the culvert headwater elevations exceed the roadway crest elevation, overtopping will begin. Calculate the upstream water surface depth above the roadway for each selected flow rate. Use these water surface depths and Equations 5.9a and 5.9b to calculate flow rates across the roadway.

Step 4 Add the culvert flow and the roadway overtopping flow at the corresponding headwater elevations to obtain the overall culvert performance curve similar to the one shown in Figure 5.16.

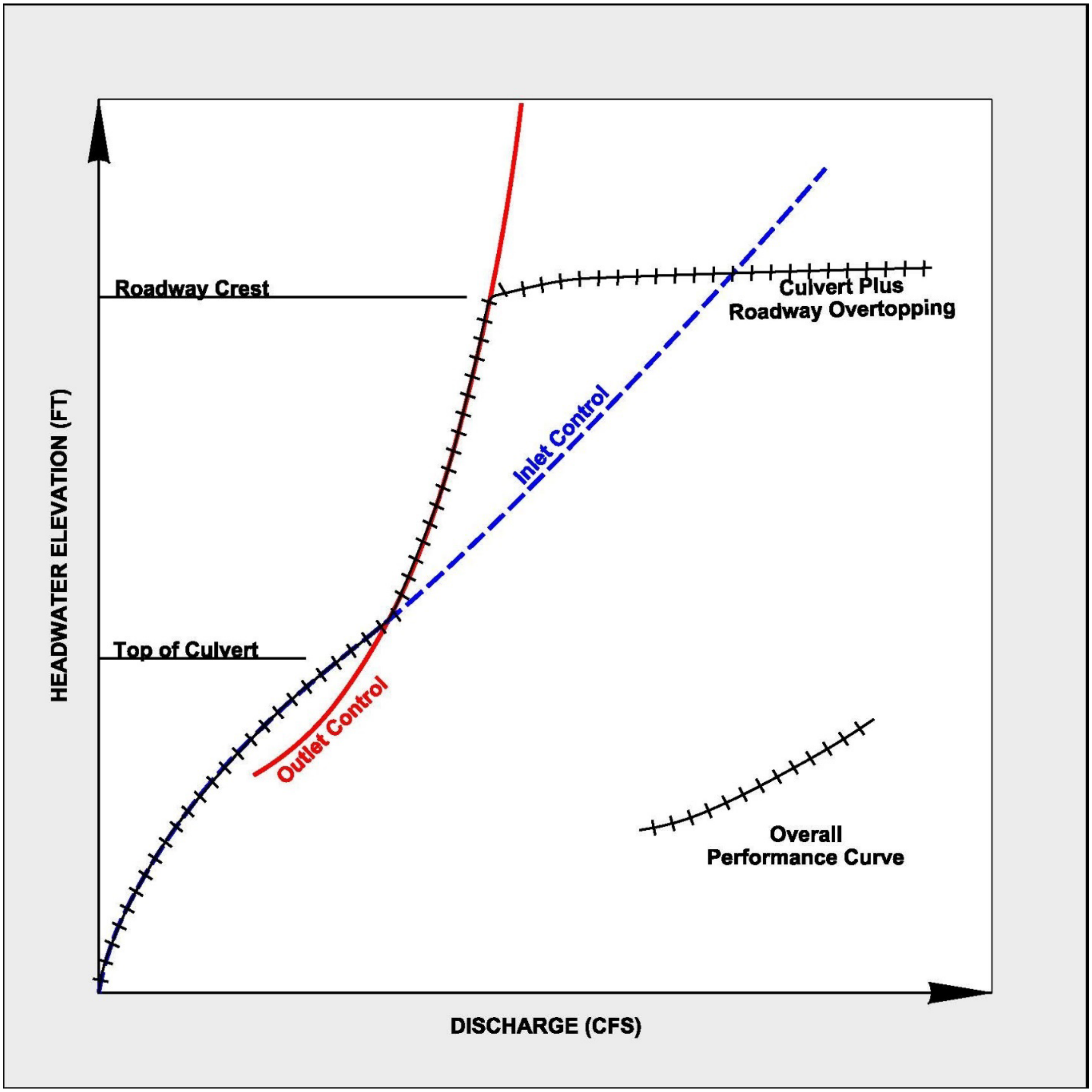


Figure 5.16 Culvert Performance Curve with Roadway Overtopping
Source: *HDS-5, Figure 3.16* (FHWA 2012)

5.4 Design Procedure

The following design procedure provides a convenient and organized method for designing culverts for a constant discharge, considering both inlet and outlet control. The procedure does not address the effect of storage which is discussed in Chapter 7.

- The designer should be familiar with all the equations in Section 5.3 before using these procedures. Following the design method without an understanding of culvert hydraulics can result in an inadequate, unsafe, or costly structure.

- The computation form has been provided in Figure 5.17 to guide the user. It contains blocks for the project description, designer's identification, hydrologic data, culvert dimensions and elevations, trial culvert description, inlet, and outlet control HW, culvert barrel selected, and comments.

- The culvert design procedure adopted by MnDOT is less rigorous for culverts with a 48" and smaller diameter or span, than for those greater than 48" where the risk assessment form is needed.

- Nomographs are available in Appendix F.

Steps 1-4 are followed when using either nomographs or computer software applications to perform culvert design computations.

Step 1. Determine Culvert Size and Scope of Culvert Design

Select initial culvert size. Culvert design is an iterative procedure where multiple culvert sizes and materials may need to be analyzed to select the best option.

Minor Culvert Designs (48" or less diameter, equivalent diameter, or box span)

- Design frequencies are provided in Table 5.1.
- Overtopping calculations are generally not necessary for the design discharge but may be computed for the check storm.
- The need for a check storm should also be considered in some situations. (See Chapter 3).
- Permit needs must be determined and may require specific flow events be analyzed and considered in the design.
- Tailwater (TW) elevation can be estimated from known condition or $(d_c+D)/2$.
- Most culvert design computer applications or nomographs are usually adequate for design.
- Specify culvert material including class or gage, if applicable.
- Determine pipe bedding requirements.
- Estimate outlet velocity using computer application or pipe flow charts.
- Design outlet protection where necessary.
- Culvert Design Form (Figure 5.17) or computer application output should be used for documentation.

Major Culvert Designs (> 48" diameter, equivalent diameter, or box span)

- Complete the Risk Assessment.
- Minimum overtopping frequency is based on Table 5.2 initially and is confirmed or modified by completing the Risk Assessment.

- Permit needs must be determined and may require specific flow events be analyzed and considered in the design.
- The Basic Flood (100-year frequency) is usually analyzed.
- Overtopping or Q_{500} analysis is necessary.
- The check storm should also be considered during design. Identify if measures can be taken to increase resilience.
- Compute tailwater (TW) (See Chapter 4).
- Water surface profile necessary for outlet control with partial full flow.
- Computer method based on FHWA recommendation to use concepts of minimum performance and control sections preferred for design (Section 5.3).
- Nomograph solution can be used as check for full flow conditions.
- Performance curve required.
- Compute outlet velocity.
- Design outlet protection where necessary.
- Specify culvert material including class or gage, if applicable.
- Documentation includes Risk Assessment, hydraulic letter, computer application output, performance curve, and material selection.

Step 2. Assemble Site Data and Project File

Site data needs will vary by project. Table 5.8 contains the suggested types of data but is not and exhaustive list of what hydraulic data is required for a specific project or site.

Table 5.8 Data Needs

Type of data	Major culverts	Minor culverts
Maps	USGS quadrangle maps Site and location maps Drainage area maps or topography	Drainage area maps or topography
Photographs	Existing and proposed structure and waterway	Aerial and land photographs
Plans	Existing and proposed Roadway plans and profiles Embankment cross-sections In-place structures	Existing and proposed Roadway plans and profiles Embankment cross-sections In-place structures
Surveys	Stream profile and cross-sections Elevations of in-place structure Elevations of flood prone property Maximum observed highwater elevations Ordinary highwater if feasible	Stream profile Channel cross-section if confined channel Flood damage potential

Type of data	Major culverts	Minor culverts
Review files	Design data for nearby structures Previous recommendations Highwater and historical flood data Maintenance problems	Previous recommendations Flooding or maintenance problems
Studies by others	Flood Insurance Studies (DNR) Watershed districts overall plan Water planning organizations Water quality studies (MPCA) Flood retention (SCS, COE, counties, watershed district) Future development (Cities & Counties) Public ditch authorities Land use	Any applicable studies Land use plans Future development plans
Environmental constraints	Boat passage (DNR) Fish migration (DNR) Wildlife passage (DNR) Domestic animal passage Wetland mitigation Protected waters (DNR) Right of way limitations Commitments in review Documents Elevation of flood prone property Permit requirements	Fish migration (DNR) Protected waters (DNR) Wetland mitigation Elevation of flood prone property Other as available Permit requirements

Step 3. Determine Hydrology

A. See Chapter 3.

- Use appropriate hydrologic method for major or minor culvert.
- Compute time of concentration using appropriate method based on site conditions and hydrologic method being used.

B. In addition, the need for a check storm should also be considered in design.

Step 4. Select Design Discharge Q_d

A. See Section 5.2.

B. Determine flood frequency from criteria.

C. Determine Q from appropriate procedure.

D. Prorate Q to each barrel if more than one.

Steps 5-13 are provided for designers using the Culvert Design Form (Figure 5.17) and manual calculations of culvert performance. If HY-8, HEC-RAS, or other suitable computer software applications are used instead, the designer should generate a report showing the software input and output values, and document factors used to justify culvert selection and design. Then continue with Step 14 of this process.

Step 5. Compute Tailwater Elevation

A. See Chapter 4.

B. Minimum data required are cross-sections, "n" values, and slope of channel to compute the rating curve for channel.

Step 6. Summarize Data on Design Form

A. See Culvert Design Form (Figure 5.17). A fillable spreadsheet version is available online.

B. Use data from steps 1-5.

Step 7. Select Design Alternative

A. See Section 5.2.4.

B. Choose trial culvert material, shape, size, and entrance type.

Step 8. Determine Inlet Control Headwater Depth (HW_i)

Use the inlet control nomograph.

A. Use the appropriate nomographs for the material types and shapes being considered.

B. Calculate headwater depth (HW_i).

- Multiply HW/D by D to obtain HW to energy grade line.
- For minor culverts neglect the approach velocity HW_i = HW.
- For major culverts in confined channels include the approach velocity
  HW_i = HW - approach velocity head.

Step 9. Determine Outlet Control Headwater Depth at Inlet (HW_{oi})

A. Compare tailwater depth calculated in Step 5 with the rise of the culvert. If tailwater (TW) ≥ culvert rise (height), nomographs will provide accurate results. Skip to step 9E. If TW < D, a computer analysis is recommended for an exact solution for major culverts; nomographs will give approximate solution for minor culverts.

B. Calculate critical depth (d_c) using pipe flow nomograph charts. Note that d_c cannot exceed D. Culvert design nomographs are available in Appendix F.

C. Calculate (d_c + D)/2.

D. Determine (h_o).
h_o = the larger of TW or (d_c + D)/2.

E. Determine (k_e).
k_e = entrance loss coefficient, Table 5.7.

F. Determine losses through the culvert barrel (H).
Use nomograph or Equation 5.3 if outside range.
Locate culvert length (L) or (L_1):
The nomographs can be used for different values of "n" by modifying the culvert length as follows:

$$L_1 = L\left(\frac{n_1}{n}\right)^2$$ **Equation 5.10**

Where:

L_1 = adjusted culvert length (ft)
L = actual culvert length (ft)
n_1 = desired Manning n value
n = Manning n value on chart (See Table 5.7)

G. Calculate outlet control headwater depth (HW_{oi}).

- If the approach velocity (V_u) and the downstream velocity (V_d) are neglected use Equation 5.11, else use Equation 5.3a, 5.3e and 5.6 to include V_u and V_d.

$$HW_{oi} = H + h_o - S_o L$$ **Equation 5.11**

- If HW_{oi} is less than 1.2D and control is outlet control:
  - the barrel may flow partly full
  - the approximate method of using the greater of tailwater or (d_c + D)/2 may not be applicable
  - backwater profile calculations should be used to check the result

– if the headwater depth falls below 0.75D, the nomograph method shall not be used for major culverts

Step 10. Determine Controlling Headwater (H_{wc})

A. Compare HW_i and HW_{oi}, use the higher.

B. Compare HW_c with allowable HW and adjust culvert size if necessary.

Step 11. Compute Discharge Over the Roadway (Q_o) (Major Culverts when Appropriate)

A. Assume the upstream depth over the roadway (HW_r), calculate length of roadway crest (L), then calculate the overtopping flowrate (Q_o). See Section 5.3.4 and Equation 5.9.

B. Calculate the flow in the culvert by using the equations in Section 5.3.2 and solving for V and then Q.

Step 12. Compute Total Discharge (Q_t)

Sum the flow over the road (Q_o) and the flow in the culvert (Q_c). This sum should equal the total flow (Q_t). If not, assume a new HW_r and make another iteration.

Step 13. Calculate Outlet Velocity (V_o) And Depth (d_n). See Section 5.3.3

A. Calculate flow depth at culvert exit using either normal depth (d_n) or the water surface profile.

B. Calculate flow area (A).

C. Calculate exit velocity (V_o) = Q/A.

If outlet control is the controlling headwater:

A. Calculate flow depth at culvert exit.

- use (d_c) if d_c > TW for minor culverts
- use (d_c + TW/2) if d_c > TW for major culverts
- use (TW) if d_c < TW < D
- use (D) if D < TW

Where:

TW = tailwater
d_c = critical depth
D = pipe diameter (height)

B. Calculate flow area at outlet (A).

C. Calculate exit velocity (V_o) = Q/A.

Steps 14 – 17 are followed when using nomographs with the Culvert Design Form (Figure 5.17) or when using HY-8, HEC-RAS, or other suitable computer software applications to do the computations and generate a report showing the software input and output values.

Step 14. Review Results

Compare alternative design with constraints and assumptions. If any of the following are exceeded, repeat Steps 4 through 13:

- The barrel must have adequate cover. (See Section 5.2.5)
- The length shall be reasonably accurate.
- The proper end section is used. (See Section 5.2.4)
- The allowable headwater is not exceeded.
- The allowable overtopping flood frequency is not exceeded.

Step 15. Plot Performance Curve (Major Culverts Only)

A. Repeat steps 4 through 13 with a range of discharges which include the design discharge (Q_{design}), 100-year discharge (Q_{100}), and overtopping discharge (Q_{OT}) or 500-year discharge (Q_{500}).

B. The check storm should also be evaluated to identify potential impacts and appropriate mitigation measures to improve resilience.

C. Use the following upper limit for discharge:

   - Q_{100} if $Q_{OT} \leq Q_{100}$
   - Q_{500} if $Q_{OT} > Q_{500}$

Step 16. Other Considerations

Consider the following options.

- Tapered inlet or larger inlet with reducer if culvert is in inlet control, very long, or has limited available headwater. (Section 5.2.4.5)
- Flow routing if a large upstream headwater pool exists (Chapter 7).
- Energy dissipators if V_o exceeds design criteria (Chapter 6).
- Sediment control storage for sites with sediment concerns such as alluvial fans.
- Fish passage or AOP design (Section 5.2.6 and consult with DNR).
- Improve resilience by using energy dissipation or erosion protection methods (Chapter 6). This may include armoring embankment slopes with riprap, vegetated riprap revetment, or permanent erosion control products.

Step 17. Documentation

- Culvert Design Form or design computations with recommendation (all Minor Culverts)
- Design computations with summary of data needed in the plan (all Major Culverts)
- Risk Assessment located in Appendix A (all Major Culverts)

- Waterway Letter of Recommendation (Use template for Major culverts that are bridges, bridge culverts or locations requiring a DNR Public Waters Permit)
- Additional information on documentation is in Section 1.4.

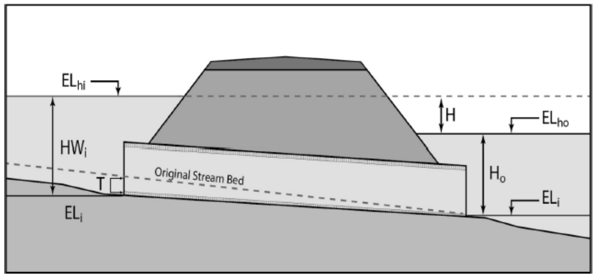
CULVERT DESIGN FORM

Project		Station		Sheet	of		Designer		Date	
							Reviewer		Date	

HYDROLOGICAL DATA	
☐ Method	
☐ Drainage area	
☐ Stream Slope	
☐ Channel shape	
☐ Routing	
☐ Other	

DESIGN FLOWS / TAILWATER		
R.I. (years)	FLOW (cfs)	TW (ft)

Roadway Elev. =
EL_{ho} =
EL_i =
$S = S_o - T/L_o$ =
L_o =
EL_o =



CULVERT DESCRIPTION	Total Flow Q (cfs)	Flow Per Barrel Q/N (1)	HEADWATER CALCULATIONS											Control Head-water Elevation	Outlet Velocity	Comments
			INLET CONTROL				OUTLET CONTROL									
Material - Shape - Size - Entrance			HW_i/D (2)	HW_i	T (3)	EL_{hi} (4)	TW (5)	d_c	$(d_c+D)/2$	H_o (6)	k_e	H (7)	EL_{ho} (8)			

TECHNICAL FOOTNOTES	
(1) Use Q/NB for box culverts	(5) TW based on downstream control or flow depth in channel
(2) $HW_i/D = HW/D$ or HW_f/D from design charts	(6) H_o = TW or $(d_c + D)/2$ (whichever is greater)
(3) $T = HW_i - (EL_{hd} - EL_{sf})$; T is zero for culverts on grade	(7) $H = [1 + k_e + (K_u n^2 L) / R^{1.33}]\, v^2/(2g)$; where K_u = 19.63 (29 in English units)
(4) $EL_{hi} = HW_i + EL_i$; (Invert of inlet control section)	(8) $EL_{ho} = EL_o + H + H_o$

SUBSCRIPT DEFINITIONS			
a	Approximate	**i**	Inlet control section
f	Culvert face	**o**	Outlet
ha	Allowable headwater	**sf**	Streambed at culvert face
hi	Headwater in inlet control	**tw**	Tailwater
ho	Headwater in outlet control		

COMMENTS / DISCUSSION

CULVERT BARREL SELECTED	
Size	
Shape	
Material	
Entrance	

Figure 5.17 Culvert Design Form
Source: *HDS-5, Figure 3.17* (FHWA 2012)

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